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MIL-M-28787/434
17 May 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY DATA MODULE, KEY CODE JML

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic memory data module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 2.138 failures per million operating hours for class I or 5.618 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be JML.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vco stress	IV	O	----------Module shall pass all remaining tests----------						
D1-D12	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D13, D14	Ibd	V	S	-0.58	0.68	-0.58	0.68	-0.58	0.68	mA
D15	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D16	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D17	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D18	Ibd	V	S	0 0	0.6	0.0	0.6	0.0	0 6	mA
D19, D20	Ibd	V	S	0 0	0.2	0 0	0.2	0.0	0 2	mA
D21	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D22	Ibd	V	S	0 0	0.2	0.0	0.2	0.0	0.2	mA
D23	Ibd	V	S	-0.58	0.68	-0.58	0.68	-0.58	0.68	mA
D24-D27	Ibd	V	S	0 0	0.1	0 0	0.1	0.0	0 1	mA
D28	Ibd	V	S	-0 58	0.68	-0 58	0.68	-0.58	0 68	mA
D29-D32	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D33	Ibd	V	S	0.0	0.5	0.0	0.5	0.0	0.5	mA
D34-D37	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D38, D39	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D40	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D41	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0 2	mA
D42	Ibd	V	S	0.0	0.5	0.0	0.5	0.0	0.5	mA
D43	Ibd	V	S	0 0	0.1	0.0	0.1	0.0	0.1	mA
D44	Ibd	V	S	0 0	0.6	0 0	0.6	0.0	0 6	mA
E1-E18	Ios	VI	S	-350 0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E19-E22	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E23	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E24-E27	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E28	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E29-E34	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E35-E38	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E39	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E40-E43	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E44	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E45-E52	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-130.0	-60.0	mA
E53,E54	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E55	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E56,E57	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E58	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E59-E62	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E63	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E64,E65	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E66	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E67,E68	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E69	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E70-E73	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E74	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E75-E77	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E78	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E79-E82	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E83	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E84-E88	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E89	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E90-E94	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E95	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E96-E98	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E99	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E100-E103	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E104	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E105,E106	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E107	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E108-E113	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E114	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E115-E119	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA

- See footnotes at end of table.

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				Min	Max	Min	Max	Min	Max	
E120	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E121-E125	Ios	VI	S	-150.0	-60.0	-150.0	-80.0	-150.0	-60.0	mA
E126	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E127-E132	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E133	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E134-E138	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E139	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
E140-E144	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E145	Ios	VI	S	-130.0	-25.0	-130.0	-25.0	-130.0	-25.0	mA
F1-F4	Ith	VII	X	-2.41	0.34	-2.41	0.34	-2.41	0.34	mA
F5,F6	Ith	VII	X	-1.71	0.24	-1.71	0.24	-1.71	0.24	mA
F7	Ith	VII	X	-2.41	0.34	-2.41	0.34	-2.41	0.34	mA
F8,F9	Ith	VII	X	-1.71	0.24	-1.71	0.24	-1.71	0.24	mA
F10	Ith	VII	X	-2.41	0.34	-2.41	0.34	-2.41	0.34	mA
F11-F22	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F23	Ith	VII	X	-1.81	0.32	-1.81	0.32	-1.81	0.32	mA
F24-F26	Ith	VII	X	-1.87	0.37	-1.87	0.37	-1.87	0.37	mA
F27	Ith	VII	X	-1.71	0.24	-1.71	0.24	-1.71	0.24	mA
F28	Ith	VII	X	-2.41	0.34	-2.41	0.34	-2.41	0.34	mA
F29	Ith	VII	X	-1.71	0.24	-1.71	0.24	-1.71	0.24	mA
F30,F31	Ith	VII	X	-27.7	-14.3	-27.7	-14.3	-27.7	-14.3	mA
F32	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F33	Ith	VII	X	-1.24	0.08	-1.24	0.08	-1.24	0.08	mA
F34	Ith	VII	X	-4.8	0.08	-4.8	0.08	-4.8	0.08	mA
F35	Ith	VII	X	-3.6	0.12	-3.6	0.12	-3.6	0.12	mA
F36,F37	Ith	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F38	Ith	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F39	Ith	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F40	Ith	VII	X	-27.7	-14.3	-27.7	-14.3	-27.7	-14.3	mA
F41-F44	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F45	Ith	VII	X	-27.7	-14.3	-27.7	-14.3	-27.7	-14.3	mA
F46-F49	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F50	Ith	VII	X	-1.27	0.35	-1.27	0.95	-1.27	0.35	mA

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F51	Ith	VII	X	-3.0	0.1	-3.0	0.1	-3.0	0.1	mA
F52,F53	Ith	VII	X	-1.87	0.37	-1.87	0.37	-1.87	0.37	mA
F54-F57	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F58	Ith	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F59,F60	Ith	VII	X	-1.87	0.37	-1.87	0.37	-1.87	0.37	mA
F61	Ith	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F62	Ith	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F63	Ith	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F64	Ith	VII	X	-3.0	0.1	-3.0	0.1	-3.0	0.1	mA
F65	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F66	Ith	VII	X	-3.6	0.12	-3.6	0.12	-3.6	0.12	mA
G1-G4	Iil	VIII	X	-2.41	0.34	-2.41	0.34	-2.41	0.34	mA
G5,G6	Iil	VIII	X	-1.71	0.24	-1.71	0.24	-1.71	0.24	mA
G7	Iil	VIII	X	-2.41	0.34	-2.41	0.34	-2.41	0.34	mA
G8,G9	Iil	VIII	X	-1.71	0.24	-1.71	0.24	-1.71	0.24	mA
G10	Iil	VIII	X	-2.41	0.34	-2.41	0.34	-2.41	0.34	mA
G11-G22	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G23	Iil	VIII	X	-1.81	0.32	-1.81	0.32	-1.81	0.32	mA
G24-G26	Iil	VIII	X	-1.87	0.37	-1.87	0.37	-1.87	0.37	mA
G27	Iil	VIII	X	-1.71	0.24	-1.71	0.24	-1.71	0.24	mA
G28	Iil	VIII	X	-2.41	0.34	-2.41	0.34	-2.41	0.34	mA
G29	Iil	VIII	X	-1.71	0.24	-1.71	0.24	-1.71	0.24	mA
G30,G31	Iil	VIII	X	-27.7	-14.3	-27.7	-14.3	-27.7	-14.3	mA
G32	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G33	Iil	VIII	X	-1.24	0.08	-1.24	0.08	-1.24	0.08	mA
G34	Iil	VIII	X	-4.8	0.08	-4.8	0.08	-4.8	0.08	mA
G35	Iil	VIII	X	-3.6	0.12	-3.6	0.12	-3.6	0.12	mA
G36,G37	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G38	Iil	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G39	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G40	Iil	VIII	X	-27.7	-14.3	-27.7	-14.3	-27.7	-14.3	mA
G41-G44	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G45	Iil	VIII	X	-27.7	-14.3	-27.7	-14.3	-27.7	-14.3	mA

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G46-G49	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G50	Iil	VIII	X	-1.27	0.35	-1.27	0.35	-1.27	0.35	mA
G51	Iil	VIII	X	-3.0	0.1	-3.0	0.1	-3.0	0.1	mA
G52,G53	Iil	VIII	X	-1.87	0.37	-1.87	0.37	-1.87	0.37	mA
G54-G57	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G58	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G59,G60	Iil	VIII	X	-1.87	0.37	-1.87	0.37	-1.87	0.37	mA
G61	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G62	Iil	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G63	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G64	Iil	VIII	X	-3.0	0.1	-3.0	0.1	-3.0	0.1	mA
G65	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G66	Iil	VIII	X	-3.6	0.12	-3.6	0.12	-3.6	0.12	mA
H1-H145	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J146	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K16	Iozh	XI	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	mA
K17,K18	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K19-K34	Iozl	XI	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	mA
K35,K36	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1-L39	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1,M2	PSFC	XIII	S	2.9	5.5	2.9	5.5	2.9	5.5	uF
N	Ioo	XIV	X	700	1680	700	1680	700	1680	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 + 5°C 60 ± 5°C		Class II 4/ -55 + 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R1-R12	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R13,R14	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R15	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R16	C1/D	XVIII	Q	16	150	16	150	16	150	pF
R17	C1/D	XVIII	Q	8	175	8	175	8	175	pF
R18	C1/D	XVIII	Q	12	150	12	150	12	150	pF
R19	C1/D	XVIII	Q	4	150	4	150	4	150	pF
R20	C1/D	XVIII	Q	2	100	2	100	2	100	pF
R21	C1/D	XVIII	Q	6	100	6	100	6	100	pF
R22	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R23	C1/D	XVIII	Q	2	125	2	125	2	125	pF
R24-R27	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R28	C1/D	XVIII	Q	2	100	2	100	2	100	pF
R29-R32	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R33	C1/D	XVIII	Q	2	125	2	125	2	125	pF
R34-R37	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R38,R39	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R40	C1/D	XVIII	Q	6	100	6	100	6	100	pF
R41	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R42	C1/D	XVIII	Q	10	150	10	150	10	150	pF
R43	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R44	C1/D	XVIII	Q	12	175	12	175	12	175	pF
R45-R48	C1/D	XVIII	Q	10	200	10	200	10	200	pF
R49,R50	C1/D	XVIII	Q	12	150	12	150	12	150	pF
R51	C1/D	XVIII	Q	10	200	10	200	10	200	pF
R52,R53	C1/D	XVIII	Q	12	150	12	150	12	150	pF
R54	C1/D	XVIII	Q	10	200	10	200	10	200	pF
R55	C1/D	XVIII	Q	16	200	16	200	16	200	pF
R56-R58	C1/D	XVIII	Q	20	200	20	200	20	200	pF
R59	C1/D	XVIII	Q	12	150	12	150	12	150	pF
R60	C1/D	XVIII	Q	10	175	10	175	10	175	pF
R61	C1/D	XVIII	Q	12	150	12	150	12	150	pF
R62	C1/D	XVIII	Q	10	200	10	200	10	200	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R63-R66	C1/a	XVIII	Q	20	200	20	200	20	200	pF
R67-R81	C1/b	XVIII	Q	4	75	4	75	4	75	pF
R82	C1/b	XVIII	Q	4	100	4	100	4	100	pF
S1-S3	Input clamp diode latch-up	XIX	S	-1.2	2.4	-1.2	2.4	-1.2	2.4	V
T1,T2	PSTR	XX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/434-1 for class I and M28787/434-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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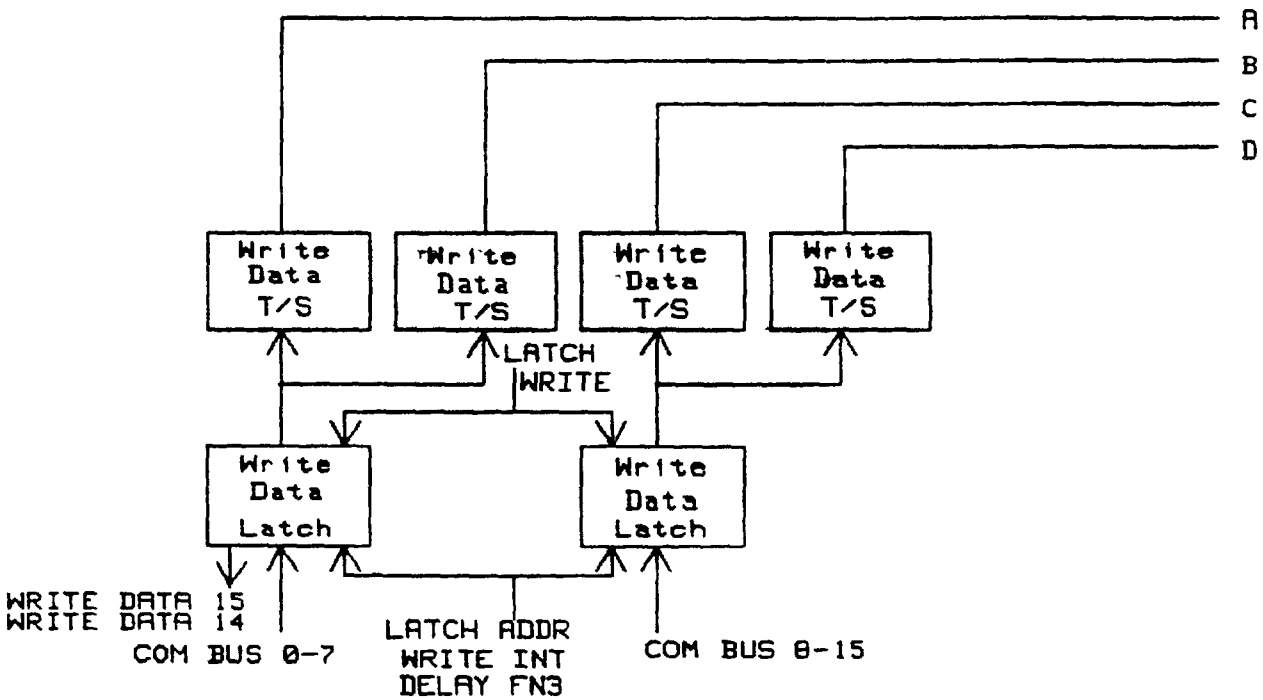


FIGURE 1. JML functional block diagram.

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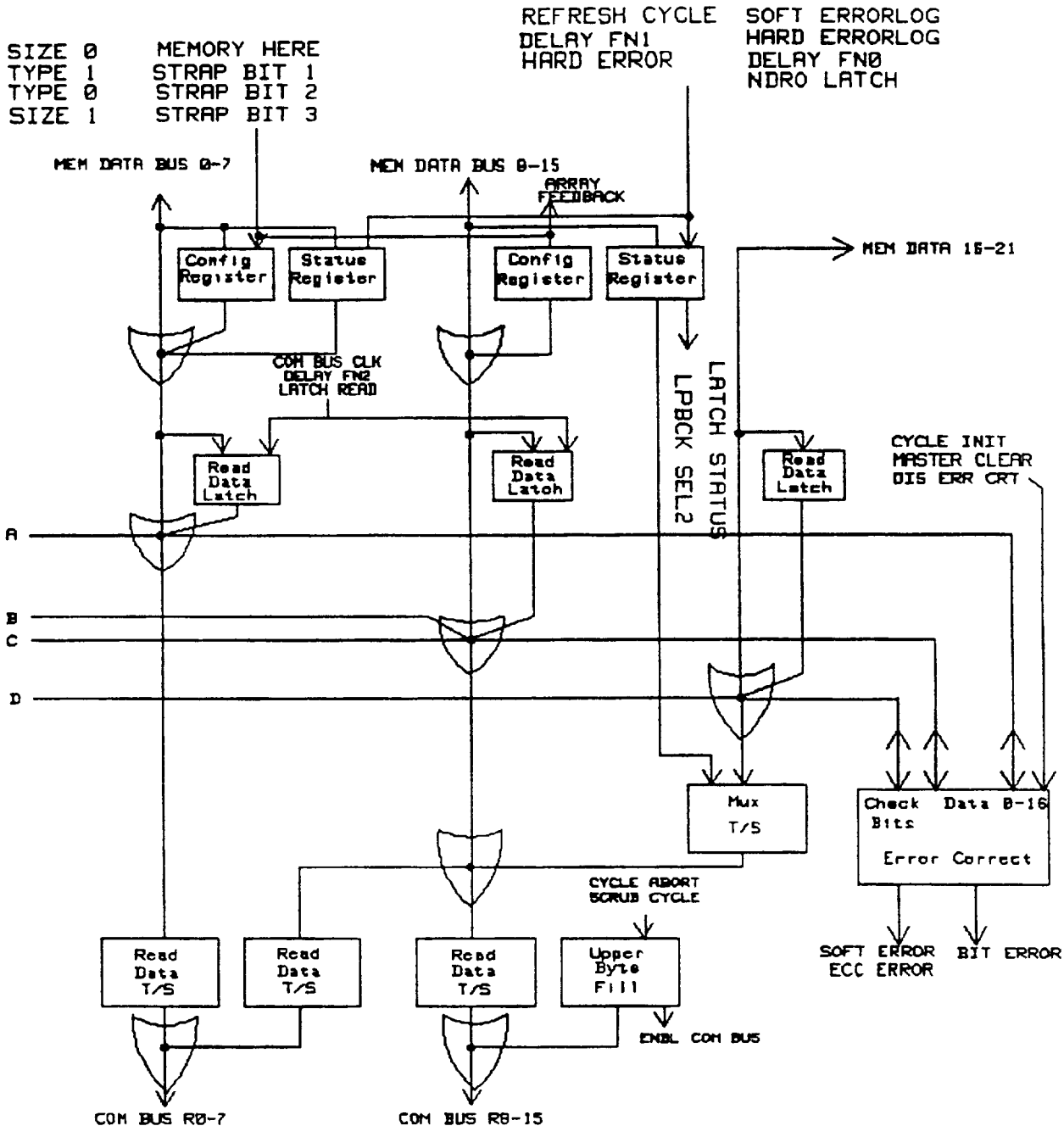


FIGURE 1. JML functional block diagram - Continued.

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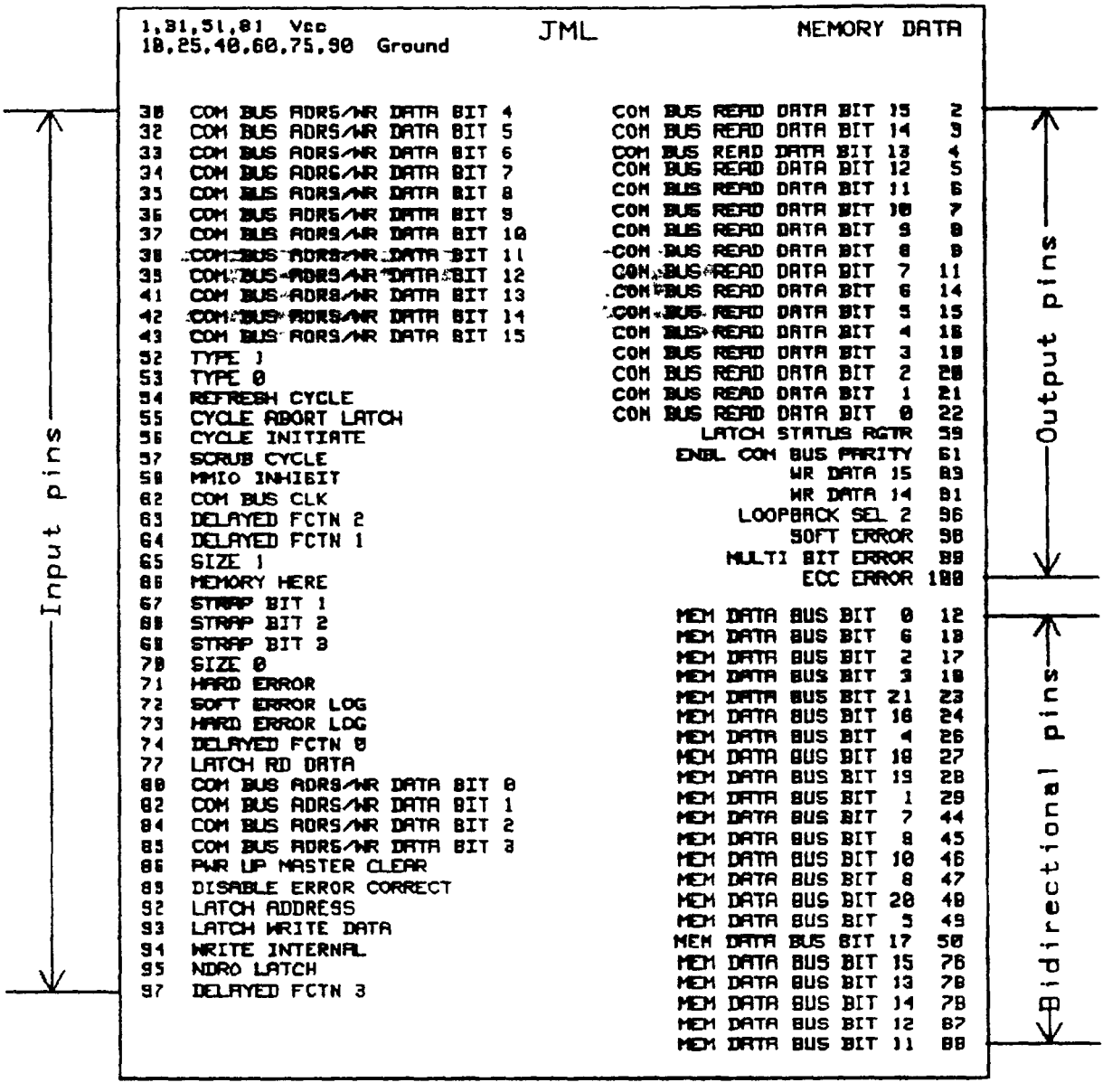


FIGURE 2. JML pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, L, and S are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, L, or S are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, L, and S where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

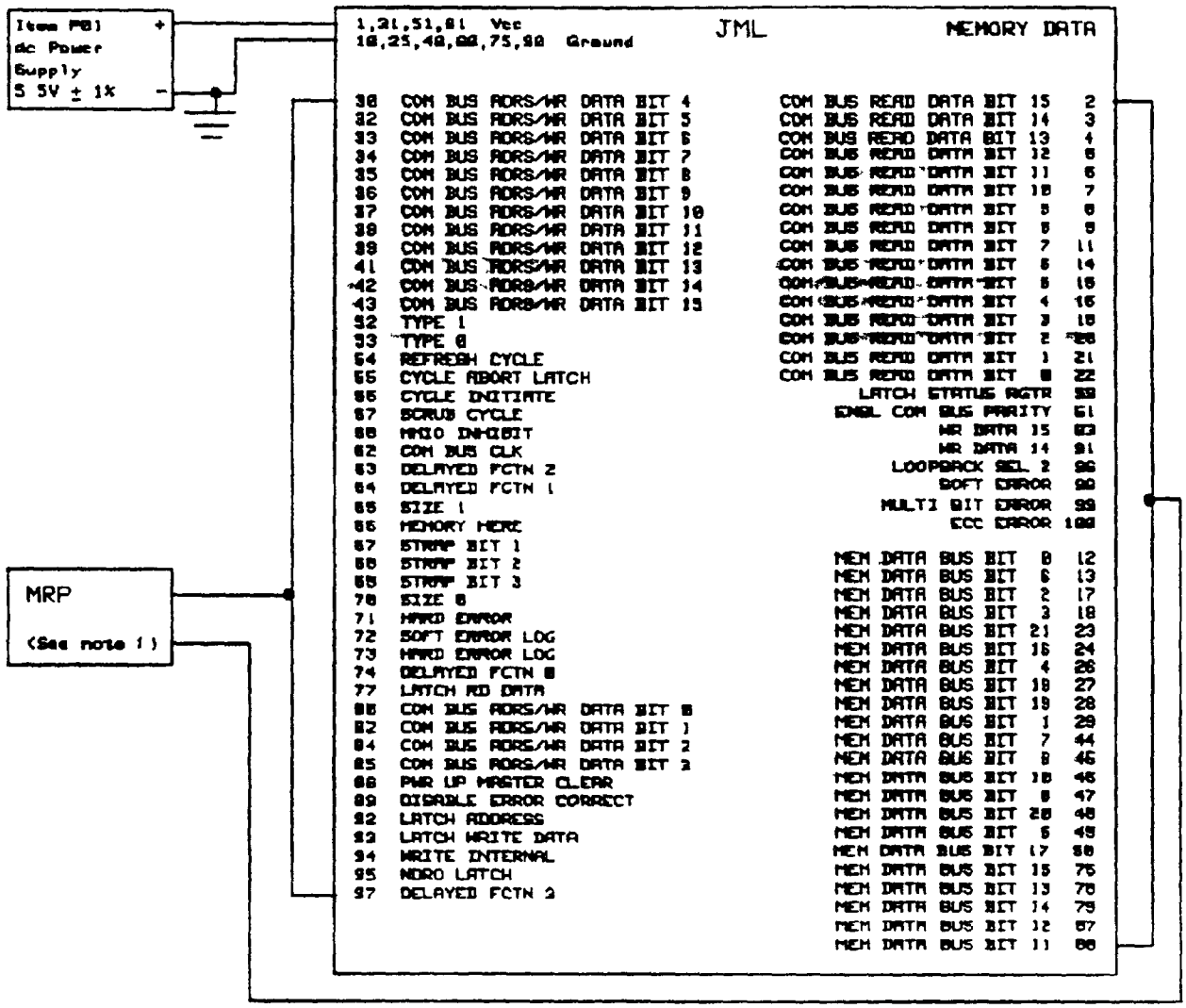
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. JML operating temperature and component temperature test circuit.

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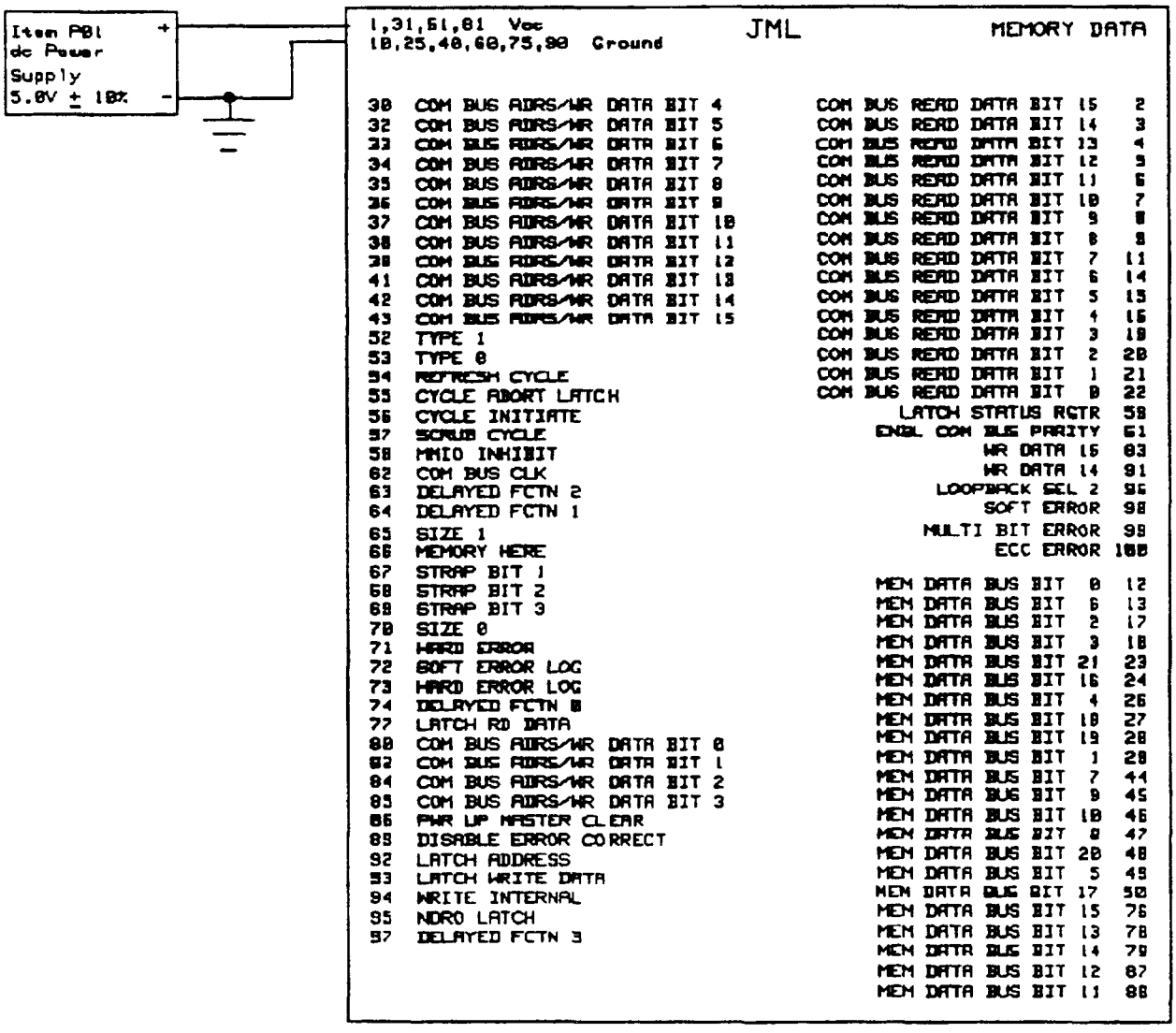


FIGURE 4. JML vibration test circuit.

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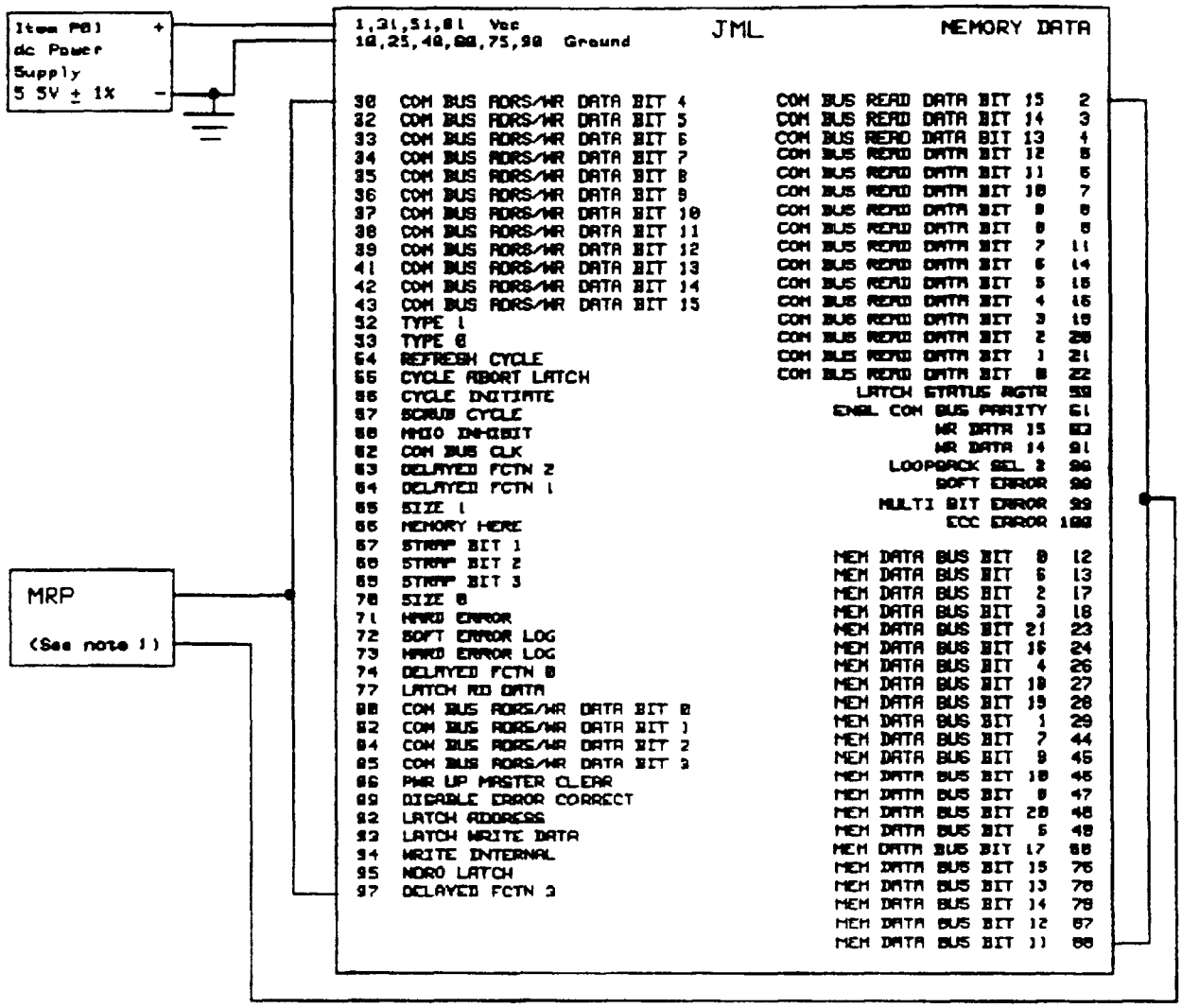
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. JML life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 9.24 W (max); 6.195 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEM DATA 0	NA	0.4	2.4	-2.41	0.34	NA	200
MEM DATA 6	NA	0.4	2.4	-2.41	0.34	NA	200
MEM DATA 2	NA	0.4	2.4	-2.41	0.34	NA	200
MEM DATA 3	NA	0.4	2.4	-2.41	0.34	NA	200
MEM DATA 21	NA	0.4	2.4	-1.71	0.24	NA	150
MEM DATA 16	NA	0.4	2.4	-1.71	0.24	NA	150
MEM DATA 4	NA	0.4	2.4	-2.41	0.34	NA	200
MEM DATA 18	NA	0.4	2.4	-1.71	0.24	NA	150
MEM DATA 19	NA	0.4	2.4	-1.71	0.24	NA	150
MEM DATA 1	NA	0.4	2.4	-2.41	0.34	NA	200
COM BUS 4	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS 5	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS 6	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS 7	-1.2	0.4	2.4	-0.6	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
COM BUS 8	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS 9	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS 10	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS 11	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS 12	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS 13	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS 14	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS 15	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MEM DATA 7	NA	0.4	2.4	-1.81	0.32	NA	200
MEM DATA 9	NA	0.4	2.4	-1.87	0.37	NA	200
MEM DATA 10	NA	0.4	2.4	-1.87	0.37	NA	200
MEM DATA 8	NA	0.4	2.4	-1.87	0.37	NA	200
MEM DATA 20	NA	0.4	2.4	-1.71	0.24	NA	150
MEM DATA 5	NA	0.4	2.4	-2.41	0.34	NA	175

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEM DATA 17	NA	0.4	2.4	-1.71	0.24	NA	150
TYPE 1	NA	0.4	2.4	-27.7	-14.3	0.68	75
TYPE 0	NA	0.4	2.4	-27.7	-14.3	0.68	75
REFRESHCYCLE	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CYCLE ABORT	NA	0.4	2.4	-1.24	0.08	1.0	150
CYCLE INIT	-1.2	0.4	2.4	-4.8	0.08	0.4	175
SCRUB CYCLE	-1.2	0.4	2.4	-3.6	0.12	0.6	150
MMIO INHIBIT	-1.2	0.4	2.4	-1.2	0.04	0.2	150
COM BUS CLK	-1.2	0.4	2.4	-1.2	0.04	0.2	100
DELAY FN2	-1.2	0.4	2.4	-1.8	0.06	0.3	100
DELAY FN1	-1.2	0.4	2.4	-1.2	0.04	0.2	75
SIZE 1	NA	0.4	2.4	-27.7	-14.3	0.68	125
MEMORY HERE	-1.2	0.4	2.4	-0.6	0.02	0.1	75
STRAP BIT 1	-1.2	0.4	2.4	-0.6	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
STRAP BIT 2	-1.2	0.4	2.4	-0.6	0.02	0.1	75
STRAP BIT 3	-1.2	0.4	2.4	-0.6	0.02	0.1	75
SIZE 0	NA	0.4	2.4	-27.7	-14.3	0.68	100
HARD ERROR	-1.2	0.4	2.4	-0.6	0.02	0.1	75
SOFTERRORLOG	-1.2	0.4	2.4	-0.6	0.02	0.1	75
HARDERRORLOG	-1.2	0.4	2.4	-0.6	0.02	0.1	75
DELAY FN0	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MEM DATA 15	NA	0.4	2.4	-1.27	0.35	NA	200
LATCH READ	-1.2	0.4	2.4	-3.0	0.1	0.5	125
MEM DATA 13	NA	0.4	2.4	-1.87	0.37	NA	200
MEM DATA 14	NA	0.4	2.4	-1.87	0.37	NA	200
COM BUS 0	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS 1	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS 2	-1.2	0.4	2.4	-0.6	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
COM BUS 3	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MASTER CLEAR	-1.2	0.4	2.4	-1.2	0.04	0.2	75
MEM DATA 12	NA	0.4	2.4	-1.87	0.37	NA	200
MEM DATA 11	NA	0.4	2.4	-1.87	0.37	NA	200
DIS ERR CRT	-1.2	0.4	2.4	-1.2	0.04	0.2	75
LATCH ADDR	-1.2	0.4	2.4	-1.8	0.06	0.3	100
LATCH WRITE	-1.2	0.4	2.4	-1.2	0.04	0.2	75
WRITE INT	-1.2	0.4	2.4	-3.0	0.1	0.5	150
NDRO LATCH	-1.2	0.4	2.4	-0.6	0.02	0.1	75
DELAY FN3	-1.2	0.4	2.4	-3.6	0.12	0.6	175

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
COM BUS R15	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R14	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R13	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R12	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R11	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R10	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R9	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R8	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R7	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R6	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R5	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R4	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R3	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R2	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
COM BUS R1	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	75
COM BUS R0	2.4	0.5	-0.4	6.0	0.02	0.02	-60.0	100
LATCH STATUS	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ENBL COM BUS	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
WR DATA 15	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
WR DATA 14	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
LPBCK SEL2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
SOFT ERROR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
BIT ERROR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ECC ERROR	2.4	0.5	-0.4	6.0	NA	NA	-25.0	NA
MEM DATA 21	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	150
MEM DATA 20	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	150
MEM DATA 19	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	150
MEM DATA 18	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	150
MEM DATA 17	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	150

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MEM DATA 16	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	150
MEM DATA 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200
MEM DATA 14	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200
MEM DATA 13	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200
MEM DATA 12	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200
MEM DATA 11	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200
MEM DATA 10	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200
MEM DATA 9	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200
MEM DATA 8	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200
MEM DATA 7	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200
MEM DATA 6	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200
MEM DATA 5	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	175
MEM DATA 4	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200
MEM DATA 3	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MEM DATA 2	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200
MEM DATA 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200
MEM DATA 0	2.4	0.5	-0.4	6.0	NA	NA	-60.0 -25.0	200

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	COM BUS CLK to COM BUS R0-15	--	--	--	51	51	51	nS	12
2	tpd	COM BUS CLK to ECC ERROR	--	--	--	39	39	39	nS	12
3	tpd	COM BUS CLK to BIT ERROR	--	--	--	59	59	59	nS	12
4	tpd	COM BUS CLK to COM BUS R0-15	--	--	--	86	86	86	nS	12
5	tsu	COM BUS 0-15 to COM BUS CLK	64	64	64	--	--	--	nS	12
6	tpd	COM BUS CLK to MEM DATA 16-21	--	--	--	97	97	97	nS	12

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6.4 Module function. The memory data module contains write (input) data latches, read (output) data latches, error correction and detection logic, the memory configuration register, memory status register and memory data control logic. The write data latches input information from the common bus and drive the memory data lines. The read data latches hold data from the memory data lines and drive the common bus. The module contains a configuration register, which is a 16-bit register initialized by the Central Processor (CP) microcode to reflect main memory configuration. The register specifies the type of array installed (i.e., read-only, non-volatile, volatile or read/write), the size of the array, the strap bits which specify the addressed array, a memory present bit, a bit which signifies the array being addressed is a Standard Common Module (SCM) on Standard Electronic Module (SEM) array, and memory data bus bits 6-0 (these bits contain a firmware written code defining the number of 64K increments of memory available). The memory status register is a 16-bit register which contains Error Detection and Correction (EDC) logic data and EDC function enables, a scrub counter clear bit, a scrub function enable bit, loopback select bits, a bit to specify that SCM on SEM memory is installed and a hardware failure detect bit. The EDC uses modified Hamming code to detect and correct single-bit errors and to detect multiple-bit errors. Error correction is enabled by bit 4 of the memory status register. EDC logic resides on the data path to and from memory and, if enabled, appends check bits to each word being written into memory and performs an error test (using these bits) on each word coming out of memory. Check bits are designated C8, C4, C2, C1, C0 and CX and occupy data bit positions 21-16. Single- and multiple-bit error flags, generated as a result of an EDC operation, are written into the memory status register, bits 1 and 0. The memory data control logic decodes delayed common bus function bits and generates read and write data latch enables and enables to the configuration and status registers. This logic also disables the status register functions during Nondestructive Readout (NDRO) and supplies control signals to the EDC logic.

6.4.1 Status multiplexer. The status multiplexer selects either status register bits 15-0 or error log and check bit information to be placed on the common bus. Status register bit 3 and 2 provide selection for the error log and check bit information. Either an enabled condition on status register bit 3 and 2 common bus function bits 3-0 specifying a status register read operation provide an output enable for the multiplexer.

6.4.2 Memory read data latches. The read data latch is a 22-bit latch which holds data read memory for application to the EDC logic.

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6.4.3 Read data/output byte swap/byte fill drivers. The read data drivers are 3-state drivers which drive the common bus read data lines. The drivers are wired and enabled to allow the following byte manipulation function:

(a) Direct transfer of memory data 15-8 to common bus 15-8.
(b) Direct transfer of memory data 7-0 to common bus 7-0.
(c) Transfer memory data 15-8 to common bus 7-0.
(d) Upper byte fill.

The upper byte to lower byte swap is performed at the byte swap driver. Upper byte fill is performed at the byte fill driver. Enables for the drivers are generated from common bus function bits 3-0 for byte operations.

6.4.4 Write/write data/input byte swap/write check bit latches. Write data latches consist of two 3-state latches which hold data from the common bus destined for memory. Data bits are latches by the Latch Write Data signal, timed from the second Com Bus OR on the memory timing module. The write, input byte swap and write check bit data latches are wired and enabled to perform the following byte-oriented functions:

(a) Transfer write data bits 15-8 to EDC logic.
(b) Direct transfer of write data bits 15-8 to memory data bits 15-8.
(c) Direct transfer of write data bits 7-0 to memory data bits 7-0.
(d) Transfer write data bits 7-0 to memory data bits 15-8.

Enables are generated by the data latch control logic control logic from common bus function bits 3-0 for byte operations.

6.4.5 Error detection and correction logic. The error detection and correction logic generates check bits on a 16-bit data word according to modified Hamming code, and corrects the data word when check bits are supplied. For 16-bit words, 6 check bits are used. The Hamming code for a 16-bit configuration is termed 16/22 code implying a 16-bit data word, appended with 6 check bits, for a 22-bit total word length. Each check bit is generated as either an exclusive OR (XOR) or exclusive NOR (XNOR) of eight of the 16 data bits. The XOR results in an even parity check while the XNOR results in an odd parity check.

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The EDC operates in three modes:

(a) Generate mode.
(b) Detect mode.
(c) Correct mode.

In the generate mode, check bits are generated that correspond to the data available at the data bus inputs to the EDC. The check bits generated are placed on the SC5-SC0 outputs of the EDC and become data bits 21-16. In detect mode, EDC examines the data available at the data bus inputs and compares it against the word's check bits. If one or more errors are detected, the Error Control Coding (ECC) Error signal goes low. If two or more errors are detected, the Multiple Errors line also goes low. If there are no errors, both lines remain high. Regardless of whether an error is detected or not, the detect mode outputs the data word and its check bits without correction. The initial portion of the correct mode functions identically to the detect mode except that the EDC is allowed to correct (by complementing) any single-bit error in the data word prior to outputting it on the data lines. If multiple errors are detected, the word is not corrected. The generate mode is selected for write operations to memory when the Initial Write (Init Wr) signal goes low. The detect/correct modes are selected on memory when the Init Wr signal goes high.

6.4.6 Function decoder and data latch control logic. The function decoder and data latch control logic are composed of gates, flip-flops and control logic necessary to generate or use the folowing signals:

(a) Latch Address (Adrs)
(b) Error
(c) Soft Error
(d) Correct
(e) Enable ECC Data
(f) Read Driver Controls
(g) Write Data Latch Controls
(h) Initial (Init) Write
(i) Latch Status Register (Rgtr)
(j) Discrete Function Control Signals
(k) Status and Configuration Rgtr Control Signals
(l) NDRO
(m) Inhibit ECC
(n) Scrub Cycle
(o) Refresh

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The discrete function control signals are generated by decoding common bus function bits 3-0 which then flag the logic necessary to complete the memory operation generated by the requestor. Additional decoding logic is provided to recognize and decode the read and write status and configuration register common bus function bit codes. Outputs from this additional decoding logic are gated with bits from the configuration and status registers to provide the necessary control signals to read from or write into these registers. The Latch Status Rgtr signal is also generated by this additional logic and gated together with the Write Internal enable from the memory timing module. The Write Internal enable specifies a write operation to the configuration register, status register or the error log. Error, Soft Error, Correct, Inhibit ECC and Enable (Enbl) ECC Data signals are all EDC logic control signals. The Error signal flags the presence of an error (single or multiple bit) detected when the EDC is operating in the detect or correct mode. The Soft Error signal is used to flag the presence of a nonrepeating, single-bit error, detected by a scrub cycle. Correct and Enbl ECC Data signals are generated by the same flip-flop and allow the EDC logic to enter the correct operating mode and enables the ECC data to the memory data lines. Inhibit ECC is from the memory timing module status register latch and is used to inhibit the EDC logic. The NDRO signal is generated by the memory control module to inform the control set that a reference to NDRO is taking place. The NDRO signal is used by the memory data module to ensure that the status register contents are not updated by a reference to NDRO. Read Driver control signals and Write Data Latch control signals are generated from decoded common bus function bits and the Latch Read Data signal which are applied to the data and enable/select inputs of the multiplexer. Both sets of control signals select one of the byte/full word enable function. The Init Write signal is used by the EDC logic to allow it to enter the generate mode and write check bits corresponding to the data word during a write cycle. This signal also provides an enable allowing Write Data Latch control signal generation. The Latch Adrs signal, timed from the Com Bus Addressed Terminator Ready (ATR) on the memory control module, allows generation of the Init Write. The Refresh and Scrub Cycle signals, from the memory control module, provide a select input to the status multiplexer. The multiplexer outputs coded enables to the memory address module via the status request latch on the memory timing module.

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6.4.7 Description of I/O pins.

Signal name	Pin number	I/O	Description
MEM DATA 0	12	I/O	MEM DATA BUS BIT 0
MEM DATA 6	13	I/O	MEM DATA BUS BIT 6
MEM DATA 2	17	I/O	MEM DATA BUS BIT 2
MEM DATA 3	18	I/O	MEM DATA BUS BIT 3
MEM DATA 21	23	I/O	MEM DATA BUS BIT 21
MEM DATA 16	24	I/O	MEM DATA BUS BIT 16
MEM DATA 4	26	I/O	MEM DATA BUS BIT 4
MEM DATA 18	27	I/O	MEM DATA BUS BIT 18
MEM DATA 19	28	I/O	MEM DATA BUS BIT 19
MEM DATA 1	29	I/O	MEM DATA BUS BIT 1
COM BUS 4	30	I	COM BUS ADRS/WR DATA BIT 4
COM BUS 5	32	I	COM BUS ADRS/WR DATA BIT 5
COM BUS 6	33	I	COM BUS ADRS/WR DATA BIT 6
COM BUS 7	34	I	COM BUS ADRS/WR DATA BIT 7
COM BUS 8	35	I	COM BUS ADRS/WR DATA BIT 8
COM BUS 9	36	I	COM BUS ADRS/WR DATA BIT 9
COM BUS 10	37	I	COM BUS ADRS/WR DATA BIT 10
COM BUS 11	38	I	COM BUS ADRS/WR DATA BIT 11
COM BUS 12	39	I	COM BUS ADRS/WR DATA BIT 12
COM BUS 13	41	I	COM BUS ADRS/WR DATA BIT 13

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6.4.7 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
COM BUS 14	42	I	COM BUS ADRS/WR DATA BIT 14
COM BUS 15	43	I	COM BUS ADRS/WR DATA BIT 15
MEM DATA 7	44	I/O	MEM DATA BUS BIT 7
MEM DATA 9	45	I/O	MEM DATA BUS BIT 9
MEM DATA 10	46	I/O	MEM DATA BUS BIT 10
MEM DATA 8	47	I/O	MEM DATA BUS BIT 8
MEM DATA 20	48	I/O	MEM DATA BUS BIT 20
MEM DATA 5	49	I/O	MEM DATA BUS BIT 5
MEM DATA 17	50	I/O	MEM DATA BUS BIT 17
TYPE 1	52	I	TYPE 1
TYPE 0	53	I	TYPE 0
REFRESHCYCLE	54	I	REFRESH CYCLE
CYCLE ABORT	55	I	CYCLE ABORT LATCH
CYCLE INIT	56	I	CYCLE INITIATE
SCRUB CYCLE	57	I	SCRUB CYCLE
MMIO INHIBIT	58	I	MMIO INHIBIT
COM BUS CLK	62	I	COM BUS CLK
DELAY FN2	63	I	DELAYED FCTN 2
DELAY FN1	64	I	DELAYED FCTN 1
SIZE 1	65	I	SIZE 1

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6.4.7 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MEMORY HERE	66	I	MEMORY HERE
STRAP BIT 1	67	I	STRAP BIT 1
STRAP BIT 2	68	I	STRAP BIT 2
STRAP BIT 3	69	I	STRAP BIT 3
SIZE 0	70	I	SIZE 0
HARD ERROR	71	I	HARD ERROR
SOFTERRORLOG	72	I	SOFT ERROR LOG
HARDERRORLOG	73	I	HARD ERROR LOG
DELAY FN0	74	I	DELAYED FCTN 0
MEM DATA 15	76	I/O	MEM DATA BUS BIT 15
LATCH READ	77	I	LATCH RD DATA
MEM DATA 13	78	I/O	MEM DATA BUS BIT 13
MEM DATA 14	79	I/O	MEM DATA BUS BIT 14
COM BUS 0	80	I	COM BUS ADRS/WR DATA BIT 0
COM BUS 1	82	I	COM BUS ADRS/WR DATA BIT 1
COM BUS 2	84	I	COM BUS ADRS/WR DATA BIT 2
COM BUS 3	85	I	COM BUS ADRS/WR DATA BIT 3
MASTER CLEAR	86	I	PWR UP MASTER CLEAR
MEM DATA 12	87	I/O	MEM DATA BUS BIT 12
MEM DATA 11	88	I/O	MEM DATA BUS BIT 11

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6.4.7 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
DIS ERR CRT	89	I	DISABLE ERROR CORRECT
LATCH ADDR	92	I	LATCH ADDRESS
LATCH WRITE	93	I	LATCH WRITE DATA
WRITE INT	94	I	WRITE INTERNAL
NDRO LATCH	95	I	NDRO LATCH
DELAY FN3	97	I	DELAYED FCTN 3
COM BUS R15	2	O	COM BUS READ DATA BIT 15
COM BUS R14	3	O	COM BUS READ DATA BIT 14
COM BUS R13	4	O	COM BUS READ DATA BIT 13
COM BUS R12	5	O	COM BUS READ DATA BIT 12
COM BUS R11	6	O	COM BUS READ DATA BIT 11
COM BUS R10	7	O	COM BUS READ DATA BIT 10
COM BUS R9	8	O	COM BUS READ DATA BIT 9
COM BUS R8	9	O	COM BUS READ DATA BIT 8
COM BUS R7	11	O	COM BUS READ DATA BIT 7
COM BUS R6	14	O	COM BUS READ DATA BIT 6
COM BUS R5	15	O	COM BUS READ DATA BIT 5
COM BUS R4	16	O	COM BUS READ DATA BIT 4
COM BUS R3	19	O	COM BUS READ DATA BIT 3
COM BUS R2	20	O	COM BUS READ DATA BIT 2

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6.4.7 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
COM BUS R1	21	O	COM BUS READ DATA BIT 1
COM BUS R0	22	O	COM BUS READ DATA BIT 0
LATCH STATUS	59	O	LATCH STATUS RGTR
ENBL COM BUS	61	O	ENBL COM BUS PARITY
WR DATA 15	83	O	WR DATA 15
WR DATA 14	91	O	WR DATA 14
LPBCK SEL2	96	O	LOOPBACK SEL 2
SOFT ERROR	98	O	SOFT ERROR
BIT ERROR	99	O	MULTI BIT ERROR
ECC ERROR	100	O	ECC ERROR

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0076)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation	C	Frame	10	INF	MOhm	1-8

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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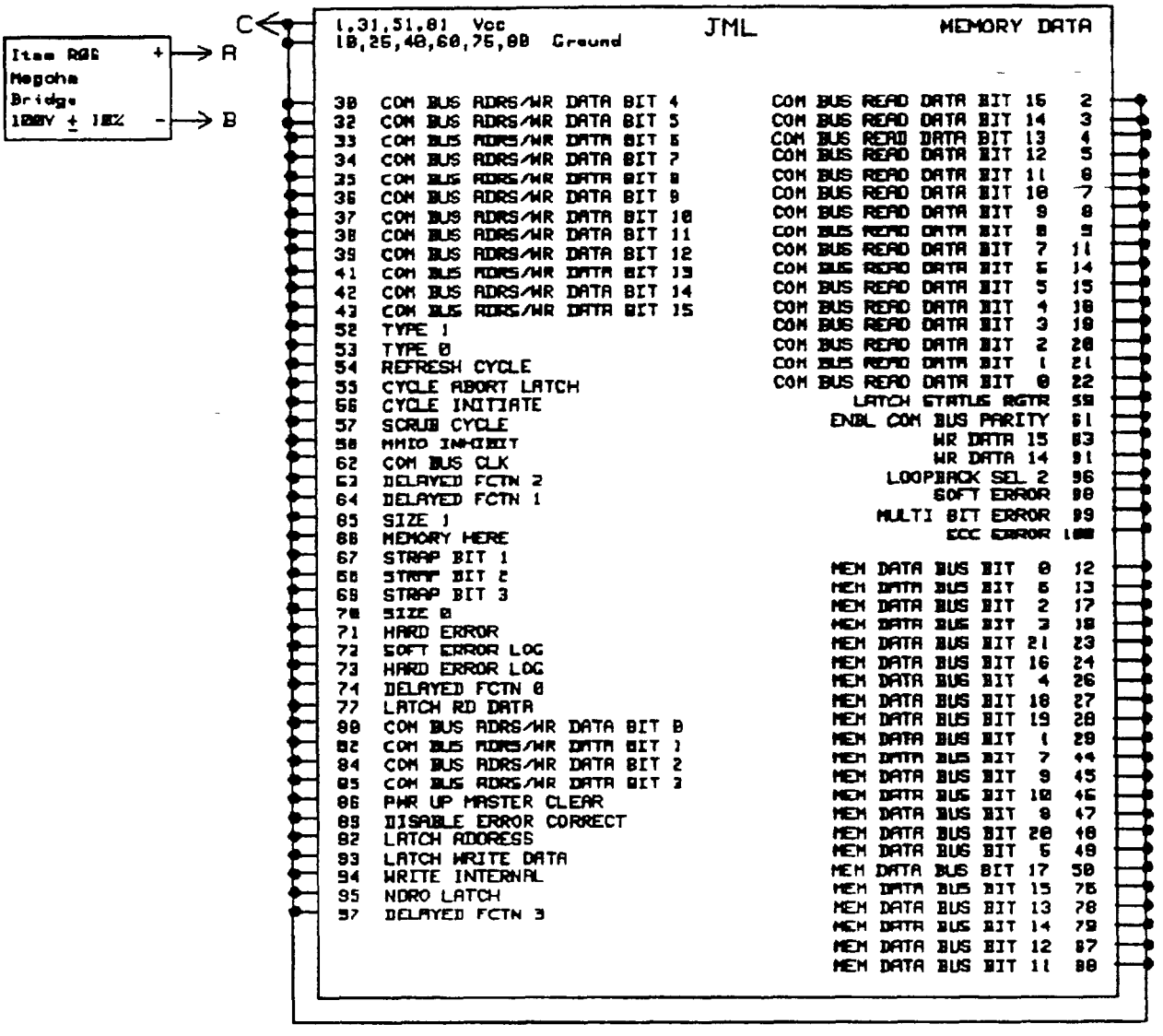


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,61,81 Vcc
10,25,40,60,75,90 Ground

JML

MEMORY DATA

30	COM BUS ADRS/WR DATA BIT 4		COM BUS READ DATA BIT 15		2
32	COM BUS ADRS/WR DATA BIT 5		COM BUS READ DATA BIT 14		3
33	COM BUS ADRS/WR DATA BIT 6		COM BUS READ DATA BIT 13		4
34	COM BUS ADRS/WR DATA BIT 7		COM BUS READ DATA BIT 12		5
35	COM BUS ADRS/WR DATA BIT 8		COM BUS READ DATA BIT 11		6
36	COM BUS ADRS/WR DATA BIT 9		COM BUS READ DATA BIT 10		7
37	COM BUS ADRS/WR DATA BIT 10		COM BUS READ DATA BIT 9		8
38	COM BUS ADRS/WR DATA BIT 11		COM BUS READ DATA BIT 8		9
39	COM BUS ADRS/WR DATA BIT 12		COM BUS READ DATA BIT 7		11
41	COM BUS ADRS/WR DATA BIT 13		COM BUS READ DATA BIT 6		14
42	COM BUS ADRS/WR DATA BIT 14		COM BUS READ DATA BIT 5		15
43	COM BUS ADRS/WR DATA BIT 15		COM BUS READ DATA BIT 4		16
52	TYPE 1		COM BUS READ DATA BIT 3		19
53	TYPE 0		COM BUS READ DATA BIT 2		20
54	REFRESH CYCLE		COM BUS READ DATA BIT 1		21
55	CYCLE ABORT LATCH		COM BUS READ DATA BIT 0		22
56	CYCLE INITIATE		LATCH STATUS RGTR		59
57	SCRUB CYCLE		ENBL COM BUS PARITY		61
60	NMIO INHIBIT		WR DATA 15		83
62	COM BUS CLK		WR DATA 14		81
63	DELAYED FCTN 2		LOOPBACK SEL 2		96
64	DELAYED FCTN 1		SOFT ERROR		98
65	SIZE 1		MULTI BIT ERROR		99
66	MEMORY HERE		ECC ERROR		100
67	STRAP BIT 1				
68	STRAP BIT 2		MEM DATA BUS BIT 0		12
69	STRAP BIT 3		MEM DATA BUS BIT 6		13
70	SIZE 0		MEM DATA BUS BIT 2		17
71	HARD ERROR		MEM DATA BUS BIT 3		18
72	SOFT ERROR LOG		MEM DATA BUS BIT 21		23
73	HARD ERROR LOG		MEM DATA BUS BIT 16		24
74	DELAYED FCTN 0		MEM DATA BUS BIT 4		26
77	LATCH RD DATA		MEM DATA BUS BIT 18		27
80	COM BUS ADRS/WR DATA BIT 0		MEM DATA BUS BIT 19		28
82	COM BUS ADRS/WR DATA BIT 1		MEM DATA BUS BIT 1		29
84	COM BUS ADRS/WR DATA BIT 2		MEM DATA BUS BIT 7		44
85	COM BUS ADRS/WR DATA BIT 3		MEM DATA BUS BIT 9		45
86	PWR UP MASTER CLEAR		MEM DATA BUS BIT 10		46
89	DISABLE ERROR CORRECT		MEM DATA BUS BIT 8		47
92	LATCH ADDRESS		MEM DATA BUS BIT 20		48
93	LATCH WRITE DATA		MEM DATA BUS BIT 5		49
94	WRITE INTERNAL		MEM DATA BUS BIT 17		50
95	MICRO LATCH		MEM DATA BUS BIT 15		76
97	DELAYED FCTN 3		MEM DATA BUS BIT 13		78
			MEM DATA BUS BIT 14		79
			MEM DATA BUS BIT 12		87
			MEM DATA BUS BIT 11		88

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	3.024 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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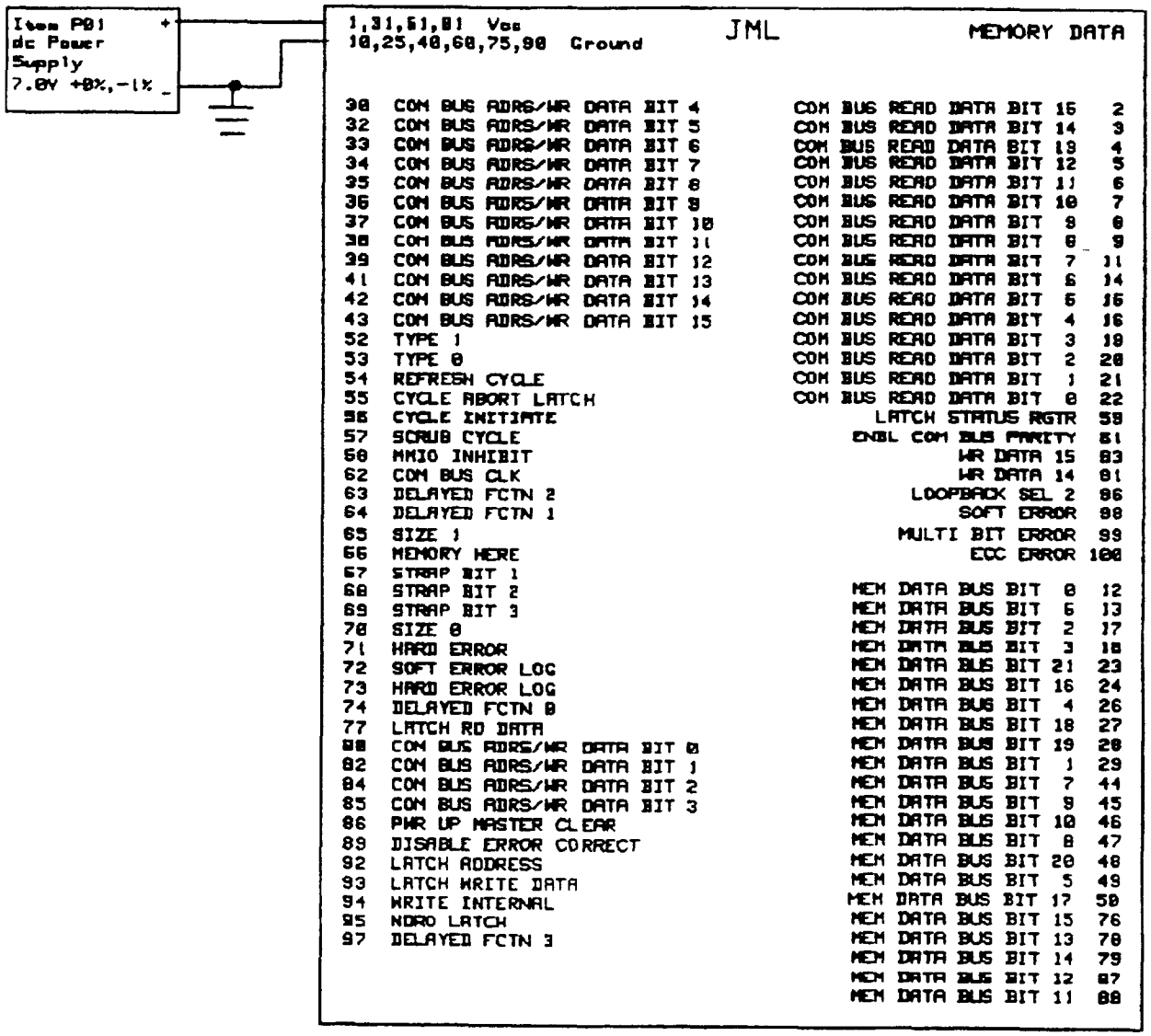


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	COM BUS 4	30	5.5	0.0	0.1	mA	1-3
D2	COM BUS 5	32					
D3	COM BUS 6	33					
D4	COM BUS 7	34					
D5	COM BUS 8	35					
D6	COM BUS 9	36					
D7	COM BUS 10	37					
D8	COM BUS 11	38					
D9	COM BUS 12	39					
D10	COM BUS 13	41					
D11	COM BUS 14	42					
D12	COM BUS 15	43	5.5	0.0	0.1	mA	1-3
D13	TYPE 1	52	5.5	-0.58	0.68	mA	1-3
D14	TYPE 0	53	5.5	-0.58	0.68	mA	1-3
D15	REFRESHCYCLE	54	5.5	0.0	0.1	mA	1-3
D16	CYCLE ABORT	55	5.5	0.0	1.0	mA	1-3
D17	CYCLE INIT	56	5.5	0.0	0.4	mA	1-3
D18	SCRUB CYCLE	57	5.5	0.0	0.6	mA	1-3
D19	MMIO INHIBIT	58	5.5	0.0	0.2	mA	1-3
D20	COM BUS CLK	62	5.5	0.0	0.2	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	DELAY FN2	63	5.5	0.0	0.3	mA	1-3
D22	DELAY FN1	64	5.5	0.0	0.2	mA	1-3
D23	SIZE 1	65	5.5	-0.58	0.68	mA	1-3
D24	MEMORY HERE	66	5.5	0.0	0.1	mA	1-3
D25	STRAP BIT 1	67					
D26	STRAP BIT 2	68					
D27	STRAP BIT 3	69	5.5	0.0	0.1	mA	1-3
D28	SIZE 0	70	5.5	-0.58	0.68	mA	1-3
D29	HARD ERROR	71	5.5	0.0	0.1	mA	1-3
D30	SOFTERRORLOG	72					
D31	HARDERRORLOG	73					
D32	DELAY FN0	74	5.5	0.0	0.1	mA	1-3
D33	LATCH READ	77	5.5	0.0	0.5	mA	1-3
D34	COM BUS 0	80	5.5	0.0	0.1	mA	1-3
D35	COM BUS 1	82					
D36	COM BUS 2	84					
D37	COM BUS 3	85	5.5	0.0	0.1	mA	1-3
D38	MASTER CLEAR	86	5.5	0.0	0.2	mA	1-3
D39	DIS ERR CRT	89	5.5	0.0	0.2	mA	1-3
D40	LATCH ADDR	92	5.5	0.0	0.3	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	LATCH WRITE	93	5.5	0.0	0.2	mA	1-3
D42	WRITE INT	94	5.5	0.0	0.5	mA	1-3
D43	NDRO LATCH	95	5.5	0.0	0.1	mA	1-3
D44	DELAY FN3	97	5.5	0.0	0.6	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D44
Power supply current limit	2.52 A	NA	D1-D44
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D44
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	D1-D44
low, (VL)	0.0 V	± 100 mV	D1-D44
Input current measurement	NA	± 5 uA	D1-D12
	NA	± 63 uA	D13,D14
	NA	± 5 uA	D15
	NA	± 50 uA	D16
	NA	± 20 uA	D17
	NA	± 30 uA	D18
	NA	± 10 uA	D19,D20
	NA	± 15 uA	D21
	NA	± 10 uA	D22
	NA	± 63 uA	D23
	NA	± 5 uA	D24-D27
	NA	± 63 uA	D28
	NA	± 5 uA	D29-D32
	NA	± 25 uA	D33
	NA	± 5 uA	D34-D37
	NA	± 10 uA	D38,D39
	NA	± 15 uA	D40
	NA	± 10 uA	D41
	NA	± 25 uA	D42
	NA	± 5 uA	D43
	NA	± 30 uA	D44

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	COM BUS R15	2	0.0	-350.0	-60.0	mA	1-4
E2	COM BUS R15	2					
E3	COM BUS R14	3					
E4	COM BUS R14	3					
E5	COM BUS R13	4					
E6	COM BUS R13	4					
E7	COM BUS R12	5					
E8	COM BUS R12	5					
E9	COM BUS R11	6					
E10	COM BUS R11	6					
E11	COM BUS R10	7					
E12	COM BUS R10	7					
E13	COM BUS R9	8					
E14	COM BUS R9	8					
E15	COM BUS R8	9					
E16	COM BUS R8	9					
E17	COM BUS R7	11					
E18	COM BUS R7	11	0.0	-350.0	-60.0	mA	1-4
E19	MEM DATA 0	12	0.0	-150.0	-60.0	mA	1-4
E20	MEM DATA 0	12	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	MEM DATA 0	12	0.0	-150.0	-60.0	mA	1-4
E22	MEM DATA 0	12	0.0	-150.0	-60.0	mA	1-4
E23	MEM DATA 0	12	0.0	-130.0	-25.0	mA	1-4
E24	MEM DATA 6	13	0.0	-150.0	-60.0	mA	1-4
E25	MEM DATA 6	13					
E26	MEM DATA 6	13					
E27	MEM DATA 6	13	0.0	-150.0	-60.0	mA	1-4
E28	MEM DATA 6	13	0.0	-130.0	-25.0	mA	1-4
E29	COM BUS R6	14	0.0	-350.0	-60.0	mA	1-4
E30	COM BUS R6	14					
E31	COM BUS R5	15					
E32	COM BUS R5	15					
E33	COM BUS R4	16					
E34	COM BUS R4	16	0.0	-350.0	-60.0	mA	1-4
E35	MEM DATA 2	17	0.0	-150.0	-60.0	mA	1-4
E36	MEM DATA 2	17					
E37	MEM DATA 2	17					
E38	MEM DATA 2	17	0.0	-150.0	-60.0	mA	1-4
E39	MEM DATA 2	17	0.0	-130.0	-25.0	mA	1-4
E40	MEM DATA 3	18	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
E41	MEM DATA 3	18	0.0	-150.0	-60.0	mA	1-4
E42	MEM DATA 3	18					
E43	MEM DATA 3	18	0.0	-150.0	-60.0	mA	1-4
E44	MEM DATA 3	18	0.0	-130.0	-25.0	mA	1-4
E45	COM BUS R3	19	0.0	-350.0	-60.0	mA	1-4
E46	COM BUS R3	19					
E47	COM BUS R2	20					
E48	COM BUS R2	20					
E49	COM BUS R1	21					
E50	COM BUS R1	21					
E51	COM BUS R0	22					
E52	COM BUS R0	22	0.0	-350.0	-60.0	mA	1-4
E53	MEM DATA 21	23	0.0	-150.0	-60.0	mA	1-4
E54	MEM DATA 21	23	0.0	-150.0	-60.0	mA	1-4
E55	MEM DATA 21	23	0.0	-130.0	-25.0	mA	1-4
E56	MEM DATA 16	24	0.0	-150.0	-60.0	mA	1-4
E57	MEM DATA 16	24	0.0	-150.0	-60.0	mA	1-4
E58	MEM DATA 16	24	0.0	-130.0	-25.0	mA	1-4
E59	MEM DATA 4	26	0.0	-150.0	-60.0	mA	1-4
E60	MEM DATA 4	26	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E61	MEM DATA 4	26	0.0	-150.0	-60.0	mA	1-4
E62	MEM DATA 4	26	0.0	-150.0	-60.0	mA	1-4
E63	MEM DATA 4	26	0.0	-130.0	-25.0	mA	1-4
E64	MEM DATA 18	27	0.0	-150.0	-60.0	mA	1-4
E65	MEM DATA 18	27	0.0	-150.0	-60.0	mA	1-4
E66	MEM DATA 18	27	0.0	-130.0	-25.0	mA	1-4
E67	MEM DATA 19	28	0.0	-150.0	-60.0	mA	1-4
E68	MEM DATA 19	28	0.0	-150.0	-60.0	mA	1-4
E69	MEM DATA 19	28	0.0	-130.0	-25.0	mA	1-4
E70	MEM DATA 1	29	0.0	-150.0	-60.0	mA	1-4
E71	MEM DATA 1	29					
E72	MEM DATA 1	29					
E73	MEM DATA 1	29	0.0	-150.0	-60.0	mA	1-4
E74	MEM DATA 1	29	0.0	-130.0	-25.0	mA	1-4
E75	MEM DATA 7	44	0.0	-150.0	-60.0	mA	1-4
E76	MEM DATA 7	44					
E77	MEM DATA 7	44	0.0	-150.0	-60.0	mA	1-4
E78	MEM DATA 7	44	0.0	-130.0	-25.0	mA	1-4
E79	MEM DATA 9	45	0.0	-150.0	-60.0	mA	1-4
E80	MEM DATA 9	45	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E81	MEM DATA 9	45	0.0	-150.0	-60.0	mA	1-4
E82	MEM DATA 9	45	0.0	-150.0	-60.0	mA	1-4
E83	MEM DATA 9	45	0.0	-130.0	-25.0	mA	1-4
E84	MEM DATA 9	45	0.0	-150.0	-60.0	mA	1-4
E85	MEM DATA 10	46					
E86	MEM DATA 10	46					
E87	MEM DATA 10	46					
E88	MEM DATA 10	46	0.0	-150.0	-60.0	mA	1-4
E89	MEM DATA 10	46	0.0	-130.0	-25.0	mA	1-4
E90	MEM DATA 10	46	0.0	-150.0	-60.0	mA	1-4
E91	MEM DATA 8	47					
E92	MEM DATA 8	47					
E93	MEM DATA 8	47					
E94	MEM DATA 8	47	0.0	-150.0	-60.0	mA	1-4
E95	MEM DATA 8	47	0.0	-130.0	-25.0	mA	1-4
E96	MEM DATA 8	47	0.0	-150.0	-60.0	mA	1-4
E97	MEM DATA 20	48					
E98	MEM DATA 20	48	0.0	-150.0	-60.0	mA	1-4
E99	MEM DATA 20	48	0.0	-130.0	-25.0	mA	1-4
E100	MEM DATA 5	49	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E101	MEM DATA 5	49	0.0	-150.0	-60.0	mA	1-4
E102	MEM DATA 5	49					
E103	MEM DATA 5	49	0.0	-150.0	-60.0	mA	1-4
E104	MEM DATA 5	49	0.0	-130.0	-25.0	mA	1-4
E105	MEM DATA 17	50	0.0	-150.0	-60.0	mA	1-4
E106	MEM DATA 17	50	0.0	-150.0	-60.0	mA	1-4
E107	MEM DATA 17	50	0.0	-130.0	-25.0	mA	1-4
E108	LATCH STATUS	59	0.0	-150.0	-60.0	mA	1-4
E109	ENBL COM BUS	61					
E110	MEM DATA 15	76					
E111	MEM DATA 15	76					
E112	MEM DATA 15	76					
E113	MEM DATA 15	76	0.0	-150.0	-60.0	mA	1-4
E114	MEM DATA 15	76	0.0	-130.0	-25.0	mA	1-4
E115	MEM DATA 15	76	0.0	-150.0	-60.0	mA	1-4
E116	MEM DATA 13	78					
E117	MEM DATA 13	78					
E118	MEM DATA 13	78					
E119	MEM DATA 13	78	0.0	-150.0	-60.0	mA	1-4
E120	MEM DATA 13	78	0.0	-130.0	-25.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E121	MEM DATA 13	78	0.0	-150.0	-60.0	mA	1-4
E122	MEM DATA 14	79					
E123	MEM DATA 14	79					
E124	MEM DATA 14	79					
E125	MEM DATA 14	79	0.0	-150.0	-60.0	mA	1-4
E126	MEM DATA 14	79	0.0	-130.0	-25.0	mA	1-4
E127	MEM DATA 14	79	0.0	-150.0	-60.0	mA	1-4
E128	WR DATA 15	83					
E129	MEM DATA 12	87					
E130	MEM DATA 12	87					
E131	MEM DATA 12	87					
E132	MEM DATA 12	87	0.0	-150.0	-60.0	mA	1-4
E133	MEM DATA 12	87	0.0	-130.0	-25.0	mA	1-4
E134	MEM DATA 12	87	0.0	-150.0	-60.0	mA	1-4
E135	MEM DATA 11	88					
E136	MEM DATA 11	88					
E137	MEM DATA 11	88					
E138	MEM DATA 11	88	0.0	-150.0	-60.0	mA	1-4
E139	MEM DATA 11	88	0.0	-130.0	-25.0	mA	1-4
E140	MEM DATA 11	88	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E141	WR DATA 14	91	0.0	-150.0	-60.0	mA	1-4
E142	LPBCK SEL2	96					
E143	SOFT ERROR	98					
E144	BIT ERROR	99	0.0	-150.0	-60.0	mA	1-4
E145	ECC ERROR	100	0.0	-130.0	-25.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E145
Power supply current limit	2.52 A	NA	E1-E145
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E145
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E145 E1-E145
Output current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 14.5 mA ± 4.5 mA ± 5.25 mA ± 4.5 mA ± 5.25 mA ± 14.5 mA ± 4.5 mA ± 5.25 mA ± 4.5 mA ± 5.25 mA ± 14.5 mA ± 4.5 mA ± 5.25 mA ± 4.5 mA ± 5.25 mA ± 4.5 mA ± 5.25 mA ± 4.5 mA ± 5.25 mA ± 4.5 mA ± 5.25 mA ± 4.5 mA	E1-E18 E19-E22 E23 E24-E27 E28 E29-E34 E35-E38 E39 E40-E43 E44 E45-E52 E53,E54 E55 E56,E57 E58 E59-E62 E63 E64,E65 E66 E67,E68 E69 E70-E73

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
	NA	± 5.25 mA	E74
	NA	± 4.5 mA	E75-E77
	NA	± 5.25 mA	E78
	NA	± 4.5 mA	E79-E82
	NA	± 5.25 mA	E83
	NA	± 4.5 mA	E84-E88
	NA	± 5.25 mA	E89
	NA	± 4.5 mA	E90-E94
	NA	± 5.25 mA	E95
	NA	± 4.5 mA	E96-E98
	NA	± 5.25 mA	E99
	NA	± 4.5 mA	E100-E103
	NA	± 5.25 mA	E104
	NA	± 4.5 mA	E105,E106
	NA	± 5.25 mA	E107
	NA	± 4.5 mA	E108-E113
	NA	± 5.25 mA	E114
	NA	± 4.5 mA	E115-E119
	NA	± 5.25 mA	E120
	NA	± 4.5 mA	E121-E125
	NA	± 5.25 mA	E126
	NA	± 4.5 mA	E127-E132
	NA	± 5.25 mA	E133
	NA	± 4.5 mA	E134-E138
	NA	± 5.25 mA	E139
	NA	± 4.5 mA	E140-E144
	NA	± 5.25 mA	E145

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	MEM DATA 0	12	2.4	-2.41	0.34	mA	1-3
F2	MEM DATA 6	13					
F3	MEM DATA 2	17					
F4	MEM DATA 3	18	2.4	-2.41	0.34	mA	1-3
F5	MEM DATA 21	23	2.4	-1.71	0.24	mA	1-3
F6	MEM DATA 16	24	2.4	-1.71	0.24	mA	1-3
F7	MEM DATA 4	26	2.4	-2.41	0.34	mA	1-3
F8	MEM DATA 18	27	2.4	-1.71	0.24	mA	1-3
F9	MEM DATA 19	28	2.4	-1.71	0.24	mA	1-3
F10	MEM DATA 1	29	2.4	-2.41	0.34	mA	1-3
F11	COM BUS 4	30	2.4	-0.6	0.02	mA	1-3
F12	COM BUS 5	32					
F13	COM BUS 6	33					
F14	COM BUS 7	34					
F15	COM BUS 8	35					
F16	COM BUS 9	36					
F17	COM BUS 10	37					
F18	COM BUS 11	38					
F19	COM BUS 12	39					
F20	COM BUS 13	41	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	COM BUS 14	42	2.4	-0.6	0.02	mA	1-3
F22	COM BUS 15	43	2.4	-0.6	0.02	mA	1-3
F23	MEM DATA 7	44	2.4	-1.81	0.32	mA	1-3
F24	MEM DATA 9	45	2.4	-1.87	0.37	mA	1-3
F25	MEM DATA 10	46					
F26	MEM DATA 8	47	2.4	-1.87	0.37	mA	1-3
F27	MEM DATA 20	48	2.4	-1.71	0.24	mA	1-3
F28	MEM DATA 5	49	2.4	-2.41	0.34	mA	1-3
F29	MEM DATA 17	50	2.4	-1.71	0.24	mA	1-3
F30	TYPE 1	52	2.4	-27.7	-14.3	mA	1-3
F31	TYPE 0	53	2.4	-27.7	-14.3	mA	1-3
F32	REFRESHCYCLE	54	2.4	-0.6	0.02	mA	1-3
F33	CYCLE ABORT	55	2.4	-1.24	0.08	mA	1-3
F34	CYCLE INIT	56	2.4	-4.8	0.08	mA	1-3
F35	SCRUB CYCLE	57	2.4	-3.6	0.12	mA	1-3
F36	MMIO INHIBIT	58	2.4	-1.2	0.04	mA	1-3
F37	COM BUS CLK	62	2.4	-1.2	0.04	mA	1-3
F38	DELAY FN2	63	2.4	-1.8	0.06	mA	1-3
F39	DELAY FN1	64	2.4	-1.2	0.04	mA	1-3
F40	SIZE 1	65	2.4	-27.7	-14.3	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	MEMORY HERE	66	2.4	-0.6	0.02	mA	1-3
F42	STRAP BIT 1	67					
F43	STRAP BIT 2	68					
F44	STRAP BIT 3	69	2.4	-0.6	0.02	mA	1-3
F45	SIZE 0	70	2.4	-27.7	-14.3	mA	1-3
F46	HARD ERROR	71	2.4	-0.6	0.02	mA	1-3
F47	SOFTERRORLOG	72					
F48	HARDERRORLOG	73					
F49	DELAY FN0	74	2.4	-0.6	0.02	mA	1-3
F50	MEM DATA 15	76	2.4	-1.27	0.35	mA	1-3
F51	LATCH READ	77	2.4	-3.0	0.1	mA	1-3
F52	MEM DATA 13	78	2.4	-1.87	0.37	mA	1-3
F53	MEM DATA 14	79	2.4	-1.87	0.37	mA	1-3
F54	COM BUS 0	80	2.4	-0.6	0.02	mA	1-3
F55	COM BUS 1	82					
F56	COM BUS 2	84					
F57	COM BUS 3	85	2.4	-0.6	0.02	mA	1-3
F58	MASTER CLEAR	86	2.4	-1.2	0.04	mA	1-3
F59	MEM DATA 12	87	2.4	-1.87	0.37	mA	1-3
F60	MEM DATA 11	88	2.4	-1.87	0.37	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F61	DIS ERR CRT	89	2.4	-1.2	0.04	mA	1-3
F62	LATCH ADDR	92	2.4	-1.8	0.06	mA	1-3
F63	LATCH WRITE	93	2.4	-1.2	0.04	mA	1-3
F64	WRITE INT	94	2.4	-3.0	0.1	mA	1-3
F65	NDRO LATCH	95	2.4	-0.6	0.02	mA	1-3
F66	DELAY FN3	97	2.4	-3.6	0.12	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F66
Power supply current limit	2.52 A	NA	F1-F66
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F66
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F66
low, (VL)	0.0 V	± 100 mV	F1-F66
Input current measurement	NA	± 137.0 uA	F1-F4
	NA	± 97.5 uA	F5,F6
	NA	± 137.0 uA	F7
	NA	± 97.5 uA	F8,F9
	NA	± 137.0 uA	F10
	NA	± 31.0 uA	F11-F22
	NA	± 106.0 uA	F23
	NA	± 112.0 uA	F24-F26
	NA	± 97.5 uA	F27
	NA	± 137.0 uA	F28
	NA	± 97.5 uA	F29
	NA	± 670.0 uA	F30,F31
	NA	± 31.0 uA	F32
	NA	± 66.0 uA	F33
	NA	± 244.0 uA	F34
	NA	± 186.0 uA	F35
	NA	± 62.0 uA	F36,F37
	NA	± 93.0 uA	F38
	NA	± 62.0 uA	F39
	NA	± 670.0 uA	F40

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
	NA	± 31.0 uA	F41-F44
	NA	± 670.0 uA	F45
	NA	± 31.0 uA	F46-F49
	NA	± 81.0 uA	F50
	NA	± 155.0 uA	F51
	NA	± 112.0 uA	F52,F53
	NA	± 31.0 uA	F54-F57
	NA	± 62.0 uA	F58
	NA	± 112.0 uA	F59,F60
	NA	± 62.0 uA	F61
	NA	± 93.0 uA	F62
	NA	± 62.0 uA	F63
	NA	± 155.0 uA	F64
	NA	± 31.0 uA	F65
	NA	± 186.0 uA	F66

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	MEM DATA 0	12	0.4	-2.41	0.34	mA	1-3
G2	MEM DATA 6	13					
G3	MEM DATA 2	17					
G4	MEM DATA 3	18	0.4	-2.41	0.34	mA	1-3
G5	MEM DATA 21	23	0.4	-1.71	0.24	mA	1-3
G6	MEM DATA 16	24	0.4	-1.71	0.24	mA	1-3
G7	MEM DATA 4	26	0.4	-2.41	0.34	mA	1-3
G8	MEM DATA 18	27	0.4	-1.71	0.24	mA	1-3
G9	MEM DATA 19	28	0.4	-1.71	0.24	mA	1-3
G10	MEM DATA 1	29	0.4	-2.41	0.34	mA	1-3
G11	COM BUS 4	30	0.4	-0.6	0.02	mA	1-3
G12	COM BUS 5	32					
G13	COM BUS 6	33					
G14	COM BUS 7	34					
G15	COM BUS 8	35					
G16	COM BUS 9	36					
G17	COM BUS 10	37					
G18	COM BUS 11	38					
G19	COM BUS 12	39					
G20	COM BUS 13	41	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	COM BUS 14	42	0.4	-0.6	0.02	mA	1-3
G22	COM BUS 15	43	0.4	-0.6	0.02	mA	1-3
G23	MEM DATA 7	44	0.4	-1.81	0.32	mA	1-3
G24	MEM DATA 9	45	0.4	-1.87	0.37	mA	1-3
G25	MEM DATA 10	46					
G26	MEM DATA 8	47	0.4	-1.87	0.37	mA	1-3
G27	MEM DATA 20	48	0.4	-1.71	0.24	mA	1-3
G28	MEM DATA 5	49	0.4	-2.41	0.34	mA	1-3
G29	MEM DATA 17	50	0.4	-1.71	0.24	mA	1-3
G30	TYPE 1	52	0.4	-27.7	-14.3	mA	1-3
G31	TYPE 0	53	0.4	-27.7	-14.3	mA	1-3
G32	REFRESHCYCLE	54	0.4	-0.6	0.02	mA	1-3
G33	CYCLE ABORT	55	0.4	-1.24	0.08	mA	1-3
G34	CYCLE INIT	56	0.4	-4.8	0.08	mA	1-3
G35	SCRUB CYCLE	57	0.4	-3.6	0.12	mA	1-3
G36	MMIO INHIBIT	58	0.4	-1.2	0.04	mA	1-3
G37	COM BUS CLK	62	0.4	-1.2	0.04	mA	1-3
G38	DELAY FN2	63	0.4	-1.8	0.06	mA	1-3
G39	DELAY FN1	64	0.4	-1.2	0.04	mA	1-3
G40	SIZE 1	65	0.4	-27.7	-14.3	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	MEMORY HERE	66	0.4	-0.6	0.02	mA	1-3
G42	STRAP BIT 1	67					
G43	STRAP BIT 2	68					
G44	STRAP BIT 3	69	0.4	-0.6	0.02	mA	1-3
G45	SIZE 0	70	0.4	-27.7	-14.3	mA	1-3
G46	HARD ERROR	71	0.4	-0.6	0.02	mA	1-3
G47	SOFTERRORLOG	72					
G48	HARDERRORLOG	73					
G49	DELAY FNO	74	0.4	-0.6	0.02	mA	1-3
G50	MEM DATA 15	76	0.4	-1.27	0.35	mA	1-3
G51	LATCH READ	77	0.4	-3.0	0.1	mA	1-3
G52	MEM DATA 13	78	0.4	-1.87	0.37	mA	1-3
G53	MEM DATA 14	79	0.4	-1.87	0.37	mA	1-3
G54	COM BUS 0	80	0.4	-0.6	0.02	mA	1-3
G55	COM BUS 1	82					
G56	COM BUS 2	84					
G57	COM BUS 3	85	0.4	-0.6	0.02	mA	1-3
G58	MASTER CLEAR	86	0.4	-1.2	0.04	mA	1-3
G59	MEM DATA 12	87	0.4	-1.87	0.37	mA	1-3
G60	MEM DATA 11	88	0.4	-1.87	0.37	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G61	DIS ERR CRT	89	0.4	-1.2	0.04	mA	1-3
G62	LATCH ADDR	92	0.4	-1.8	0.06	mA	1-3
G63	LATCH WRITE	93	0.4	-1.2	0.04	mA	1-3
G64	WRITE INT	94	0.4	-3.0	0.1	mA	1-3
G65	NDRO LATCH	95	0.4	-0.6	0.02	mA	1-3
G66	DELAY FN3	97	0.4	-3.6	0.12	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G66
Power supply current limit	2.52 A	NA	G1-G66
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G66
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 G1-G66 G1-G66
Input current measurement	NA	± 137.0 uA	G1-G4
	NA	± 97.5 uA	G5,G6
	NA	± 137.0 uA	G7
	NA	± 97.5 uA	G8,G9
	NA	± 137.0 uA	G10
	NA	± 31.0 uA	G11-G22
	NA	± 106.0 uA	G23
	NA	± 112.0 uA	G24-G26
	NA	± 97.5 uA	G27
	NA	± 137.0 uA	G28
	NA	± 97.5 uA	G29
	NA	± 670.0 uA	G30,G31
	NA	± 31.0 uA	G32
	NA	± 66.0 uA	G33
	NA	± 244.0 uA	G34
	NA	± 186.0 uA	G35
	NA	± 62.0 uA	G36,G37
	NA	± 93.0 uA	G38
	NA	± 62.0 uA	G39
	NA	± 670.0 uA	G40
	NA	± 31.0 uA	G41-G44
	NA	± 670.0 uA	G45

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
	NA	± 31.0 uA	G46-G49
	NA	± 81.0 uA	G50
	NA	± 155.0 uA	G51
	NA	± 112.0 uA	G52,G53
	NA	± 31.0 uA	G54-G57
	NA	± 62.0 uA	G58
	NA	± 112.0 uA	G59,G60
	NA	± 62.0 uA	G61
	NA	± 93.0 uA	G62
	NA	± 62.0 uA	G63
	NA	± 155.0 uA	G64
	NA	± 31.0 uA	G65
	NA	± 186.0 uA	G66

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	COM BUS R15	2	-0.4	2.4	4.5	V	1-4
H2	COM BUS R15	2					
H3	COM BUS R14	3					
H4	COM BUS R14	3					
H5	COM BUS R13	4					
H6	COM BUS R13	4					
H7	COM BUS R12	5					
H8	COM BUS R12	5					
H9	COM BUS R11	6					
H10	COM BUS R11	6					
H11	COM BUS R10	7					
H12	COM BUS R10	7					
H13	COM BUS R9	8					
H14	COM BUS R9	8					
H15	COM BUS R8	9					
H16	COM BUS R8	9					
H17	COM BUS R7	11					
H18	COM BUS R7	11					
H19	MEM DATA 0	12					
H20	MEM DATA 0	12	-0.4	2.4	4.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	MEM DATA 0	12	-0.4	2.4	4.5	V	1-4
H22	MEM DATA 0	12					
H23	MEM DATA 0	12					
H24	MEM DATA 6	13					
H25	MEM DATA 6	13					
H26	MEM DATA 6	13					
H27	MEM DATA 6	13					
H28	MEM DATA 6	13					
H29	COM BUS R6	14					
H30	COM BUS R6	14					
H31	COM BUS R5	15					
H32	COM BUS R5	15					
H33	COM BUS R4	16					
H34	COM BUS R4	16					
H35	MEM DATA 2	17					
H36	MEM DATA 2	17					
H37	MEM DATA 2	17					
H38	MEM DATA 2	17					
H39	MEM DATA 2	17					
H40	MEM DATA 3	18	-0.4	2.4	4.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	MEM DATA 3	18	-0.4	2.4	4.5	V	1-4
H42	MEM DATA 3	18					
H43	MEM DATA 3	18					
H44	MEM DATA 3	18					
H45	COM BUS R3	19					
H46	COM BUS R3	19					
H47	COM BUS R2	20					
H48	COM BUS R2	20					
H49	COM BUS R1	21					
H50	COM BUS R1	21					
H51	COM BUS R0	22					
H52	COM BUS R0	22					
H53	MEM DATA 21	23					
H54	MEM DATA 21	23					
H55	MEM DATA 21	23					
H56	MEM DATA 16	24					
H57	MEM DATA 16	24					
H58	MEM DATA 16	24					
H59	MEM DATA 4	26					
H60	MEM DATA 4	26	-0.4	2.4	4.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H61	MEM DATA 4	26	-0.4	2.4	4.5	V	1-4
H62	MEM DATA 4	26					
H63	MEM DATA 4	26					
H64	MEM DATA 18	27					
H65	MEM DATA 18	27					
H66	MEM DATA 18	27					
H67	MEM DATA 19	28					
H68	MEM DATA 19	28					
H69	MEM DATA 19	28					
H70	MEM DATA 1	29					
H71	MEM DATA 1	29					
H72	MEM DATA 1	29					
H73	MEM DATA 1	29					
H74	MEM DATA 1	29					
H75	MEM DATA 7	44					
H76	MEM DATA 7	44					
H77	MEM DATA 7	44					
H78	MEM DATA 7	44					
H79	MEM DATA 9	45					
H80	MEM DATA 9	45	-0.4	2.4	4.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H81	MEM DATA 9	45	-0.4	2.4	4.5	V	1-4
H82	MEM DATA 9	45					
H83	MEM DATA 9	45					
H84	MEM DATA 9	45					
H85	MEM DATA 10	46					
H86	MEM DATA 10	46					
H87	MEM DATA 10	46					
H88	MEM DATA 10	46					
H89	MEM DATA 10	46					
H90	MEM DATA 10	46					
H91	MEM DATA 8	47					
H92	MEM DATA 8	47					
H93	MEM DATA 8	47					
H94	MEM DATA 8	47					
H95	MEM DATA 8	47					
H96	MEM DATA 8	47					
H97	MEM DATA 20	48					
H98	MEM DATA 20	48					
H99	MEM DATA 20	48					
H100	MEM DATA 5	49	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H101	MEM DATA 5	49	-0.4	2.4	4.5	V	1-4
H102	MEM DATA 5	49					
H103	MEM DATA 5	49					
H104	MEM DATA 5	49					
H105	MEM DATA 17	50					
H106	MEM DATA 17	50					
H107	MEM DATA 17	50					
H108	LATCH STATUS	59					
H109	ENBL COM BUS	61					
H110	MEM DATA 15	76					
H111	MEM DATA 15	76					
H112	MEM DATA 15	76					
H113	MEM DATA 15	76					
H114	MEM DATA 15	76					
H115	MEM DATA 15	76					
H116	MEM DATA 13	78					
H117	MEM DATA 13	78					
H118	MEM DATA 13	78					
H119	MEM DATA 13	78					
H120	MEM DATA 13	78	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H121	MEM DATA 13	78	-0.4	2.4	4.5	V	1-4
H122	MEM DATA 14	79					
H123	MEM DATA 14	79					
H124	MEM DATA 14	79					
H125	MEM DATA 14	79					
H126	MEM DATA 14	79					
H127	MEM DATA 14	79					
H128	WR DATA 15	83					
H129	MEM DATA 12	87					
H130	MEM DATA 12	87					
H131	MEM DATA 12	87					
H132	MEM DATA 12	87					
H133	MEM DATA 12	87					
H134	MEM DATA 12	87					
H135	MEM DATA 11	88					
H136	MEM DATA 11	88					
H137	MEM DATA 11	88					
H138	MEM DATA 11	88					
H139	MEM DATA 11	88					
H140	MEM DATA 11	88	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H141	WR DATA 14	91	-0.4	2.4	4.5	V	1-4
H142	LPBCK SEL2	96					
H143	SOFT ERROR	98					
H144	BIT ERROR	99					
H145	ECC ERROR	100	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H145
Power supply current limit	2.52 A	NA	H1-H145
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1-H145
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H145 H1-H145
Output voltage measurement	NA	± 105 mV	H1-H145

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

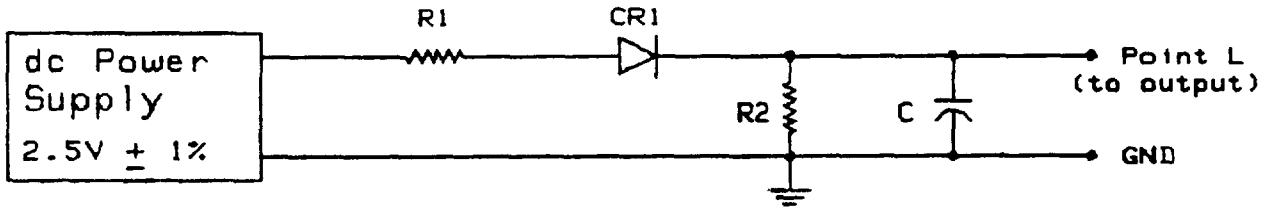
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,3,4,5,6,7,8,9, 11,12,13,14,15, 16,17,18,19,20, 21,22,23,24,26, 27,28,29,44,45, 46,47,48,49,50, 59,61,76,78,79, 83,87,88,91,96, 98,99,100	6200	60	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	COM BUS R15	2	6.0	0.0	0.5	V	1-4
J2	COM BUS R15	2					
J3	COM BUS R14	3					
J4	COM BUS R14	3					
J5	COM BUS R13	4					
J6	COM BUS R13	4					
J7	COM BUS R12	5					
J8	COM BUS R12	5					
J9	COM BUS R11	6					
J10	COM BUS R11	6					
J11	COM BUS R10	7					
J12	COM BUS R10	7					
J13	COM BUS R9	8					
J14	COM BUS R9	8					
J15	COM BUS R8	9					
J16	COM BUS R8	9					
J17	COM BUS R7	11					
J18	COM BUS R7	11					
J19	MEM DATA 0	12					
J20	MEM DATA 0	12	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	MEM DATA 0	12	6.0	0.0	0.5	V	1-4
J22	MEM DATA 0	12					
J23	MEM DATA 0	12					
J24	MEM DATA 6	13					
J25	MEM DATA 6	13					
J26	MEM DATA 6	13					
J27	MEM DATA 6	13					
J28	MEM DATA 6	13					
J29	COM BUS R6	14					
J30	COM BUS R6	14					
J31	COM BUS R5	15					
J32	COM BUS R5	15					
J33	COM BUS R4	16					
J34	COM BUS R4	16					
J35	MEM DATA 2	17					
J36	MEM DATA 2	17					
J37	MEM DATA 2	17					
J38	MEM DATA 2	17					
J39	MEM DATA 2	17					
J40	MEM DATA 3	18	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J41	MEM DATA 3	18	6.0	0.0	0.5	V	1-4
J42	MEM DATA 3	18					
J43	MEM DATA 3	18					
J44	MEM DATA 3	18					
J45	COM BUS R3	19					
J46	COM BUS R3	19					
J47	COM BUS R2	20					
J48	COM BUS R2	20					
J49	COM BUS R1	21					
J50	COM BUS R1	21					
J51	COM BUS R0	22					
J52	COM BUS R0	22					
J53	MEM DATA 21	23					
J54	MEM DATA 21	23					
J55	MEM DATA 21	23					
J56	MEM DATA 16	24					
J57	MEM DATA 16	24					
J58	MEM DATA 16	24					
J59	MEM DATA 4	26					
J60	MEM DATA 4	26	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J61	MEM DATA 4	26	6.0	0.0	0.5	V	1-4
J62	MEM DATA 4	26					
J63	MEM DATA 4	26					
J64	MEM DATA 18	27					
J65	MEM DATA 18	27					
J66	MEM DATA 18	27					
J67	MEM DATA 19	28					
J68	MEM DATA 19	28					
J69	MEM DATA 19	28					
J70	MEM DATA 1	29					
J71	MEM DATA 1	29					
J72	MEM DATA 1	29					
J73	MEM DATA 1	29					
J74	MEM DATA 1	29					
J75	MEM DATA 7	44					
J76	MEM DATA 7	44					
J77	MEM DATA 7	44					
J78	MEM DATA 7	44					
J79	MEM DATA 7	44					
J80	MEM DATA 9	45	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J81	MEM DATA 9	45	6.0	0.0	0.5	V	1-4
J82	MEM DATA 9	45					
J83	MEM DATA 9	45					
J84	MEM DATA 9	45					
J85	MEM DATA 9	45					
J86	MEM DATA 10	46					
J87	MEM DATA 10	46					
J88	MEM DATA 10	46					
J89	MEM DATA 10	46					
J90	MEM DATA 10	46					
J91	MEM DATA 10	46					
J92	MEM DATA 8	47					
J93	MEM DATA 8	47					
J94	MEM DATA 8	47					
J95	MEM DATA 8	47					
J96	MEM DATA 8	47					
J97	MEM DATA 8	47					
J98	MEM DATA 20	48					
J99	MEM DATA 20	48					
J100	MEM DATA 20	48	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J101	MEM DATA 5	49	6.0	0.0	0.5	V	1-4
J102	MEM DATA 5	49					
J103	MEM DATA 5	49					
J104	MEM DATA 5	49					
J105	MEM DATA 5	49					
J106	MEM DATA 17	50					
J107	MEM DATA 17	50					
J108	MEM DATA 17	50					
J109	LATCH STATUS	59					
J110	ENBL COM BUS	61					
J111	MEM DATA 15	76					
J112	MEM DATA 15	76					
J113	MEM DATA 15	76					
J114	MEM DATA 15	76					
J115	MEM DATA 15	76					
J116	MEM DATA 15	76					
J117	MEM DATA 13	78					
J118	MEM DATA 13	78					
J119	MEM DATA 13	78					
J120	MEM DATA 13	78	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J121	MEM DATA 13	78	6.0	0.0	0.5	V	1-4
J122	MEM DATA 13	78					
J123	MEM DATA 14	79					
J124	MEM DATA 14	79					
J125	MEM DATA 14	79					
J126	MEM DATA 14	79					
J127	MEM DATA 14	79					
J128	MEM DATA 14	79					
J129	WR DATA 15	83					
J130	MEM DATA 12	87					
J131	MEM DATA 12	87					
J132	MEM DATA 12	87					
J133	MEM DATA 12	87					
J134	MEM DATA 12	87					
J135	MEM DATA 12	87					
J136	MEM DATA 11	88					
J137	MEM DATA 11	88					
J138	MEM DATA 11	88					
J139	MEM DATA 11	88					
J140	MEM DATA 11	88	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J141	MEM DATA 11	88	6.0	0.0	0.5	V	1-4
J142	WR DATA 14	91					
J143	LPBCK SEL2	96					
J144	SOFT ERROR	98					
J145	BIT ERROR	99					
J146	ECC ERROR	⊥00	6.0	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J146
Power supply current limit	2.52 A	NA	J1-J146
Forcing current, (Iol)	6.0 mA	± 600 uA	J1-J146
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J146 J1-J146
Output voltage measurement	NA	± 25 mV	J1-J146

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	COM BUS R15	2	2.4	-0.02	0.02	mA	1-3
K2	COM BUS R14	3					
K3	COM BUS R13	4					
K4	COM BUS R12	5					
K5	COM BUS R11	6					
K6	COM BUS R10	7					
K7	COM BUS R9	8					
K8	COM BUS R8	9					
K9	COM BUS R7	11					
K10	COM BUS R6	14					
K11	COM BUS R5	15					
K12	COM BUS R4	16					
K13	COM BUS R3	19					
K14	COM BUS R2	20					
K15	COM BUS R1	21					
K16	COM BUS R0	22	2.4	-0.02	0.02	mA	1-3
K17	WR DATA 15	83	2.4	-0.05	0.05	mA	1-3
K18	WR DATA 14	91	2.4	-0.05	0.05	mA	1-3
Low level							
K19	COM BUS R15	2	0.4	-0.02	0.02	mA	1-3

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TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K20	COM BUS R14	3	0.4	-0.02	0.02	mA	1-3
K21	COM BUS R13	4					
K22	COM BUS R12	5					
K23	COM BUS R11	6					
K24	COM BUS R10	7					
K25	COM BUS R9	8					
K26	COM BUS R8	9					
K27	COM BUS R7	11					
K28	COM BUS R6	14					
K29	COM BUS R5	15					
K30	COM BUS R4	16					
K31	COM BUS R3	19					
K32	COM BUS R2	20					
K33	COM BUS R1	21					
K34	COM BUS R0	22	0.4	-0.02	0.02	mA	1-3
K35	WR DATA 15	83	0.4	-0.05	0.05	mA	1-3
K36	WR DATA 14	91	0.4	-0.05	0.05	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K36
Power supply current limit	2.52 A	NA	K1-K36
Forcing voltage, (Voh) (Vol)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 K1-K18 K19-K36
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K36 K1-K36
Output current measurement	NA NA NA NA	± 2 uA ± 5 uA ± 2 uA ± 5 uA	K1-K16 K17,K18 K19-K34 K35,K36

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	COM BUS 4	30	-18.0	-1.2	0.0	V	1-3
L2	COM BUS 5	32					
L3	COM BUS 6	33					
L4	COM BUS 7	34					
L5	COM BUS 8	35					
L6	COM BUS 9	36					
L7	COM BUS 10	37					
L8	COM BUS 11	38					
L9	COM BUS 12	39					
L10	COM BUS 13	41					
L11	COM BUS 14	42					
L12	COM BUS 15	43					
L13	REFRESHCYCLE	54					
L14	CYCLE INIT	56					
L15	SCRUB CYCLE	57					
L16	MMIO INHIBIT	58					
L17	COM BUS CLK	62					
L18	DELAY FN2	63					
L19	DELAY FN1	64					
L20	MEMORY HERE	66	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	STRAP BIT 1	67	-18.0	-1.2	0.0	V	1-3
L22	STRAP BIT 2	68					
L23	STRAP BIT 3	69					
L24	HARD ERROR	71					
L25	SOFTERRORLOG	72					
L26	HARDERRORLOG	73					
L27	DELAY FN0	74					
L28	LATCH READ	77					
L29	COM BUS 0	80					
L30	COM BUS 1	82					
L31	COM BUS 2	84					
L32	COM BUS 3	85					
L33	MASTER CLEAR	86					
L34	DIS ERR CRT	89					
L35	LATCH ADDR	92					
L36	LATCH WRITE	93					
L37	WRITE INT	94					
L38	NDRO LATCH	95					
L39	DELAY FN3	97	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L39
Power supply current limit	NA	NA	L1-L39
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L39
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L39 L1-L39
Input voltage measurement	NA	± 60 mV	L1-L39

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	2.9	5.5	uF	1-4
M2	PSFC	51	2.9	5.5	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA	± 130 nF	M1,M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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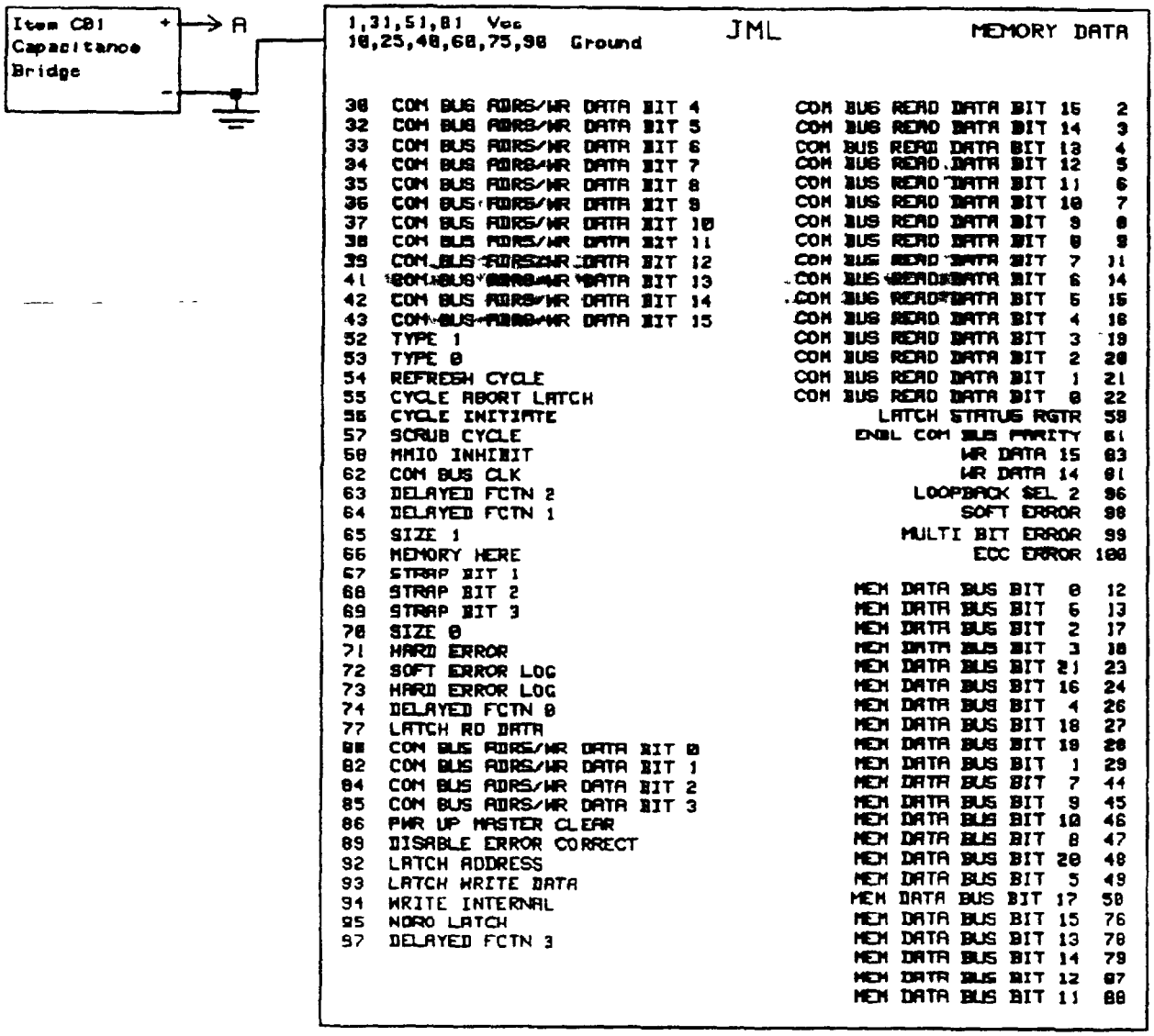


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31 51,81	700	1680	mA	1-3

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NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	2.52 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 49 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (400 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (400 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	2.52 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

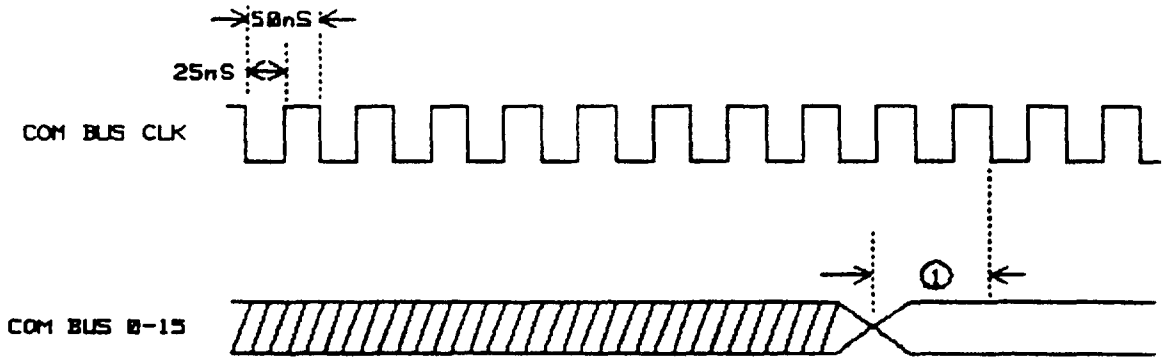
(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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NOTES:

1. (1) COM BUS 0-15 to COM BUS CLK.

   Setup time: 64 nS minimum.

FIGURE 11. Input timing parameters for table XV.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (400 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (400 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (400 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (400 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	2.52 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

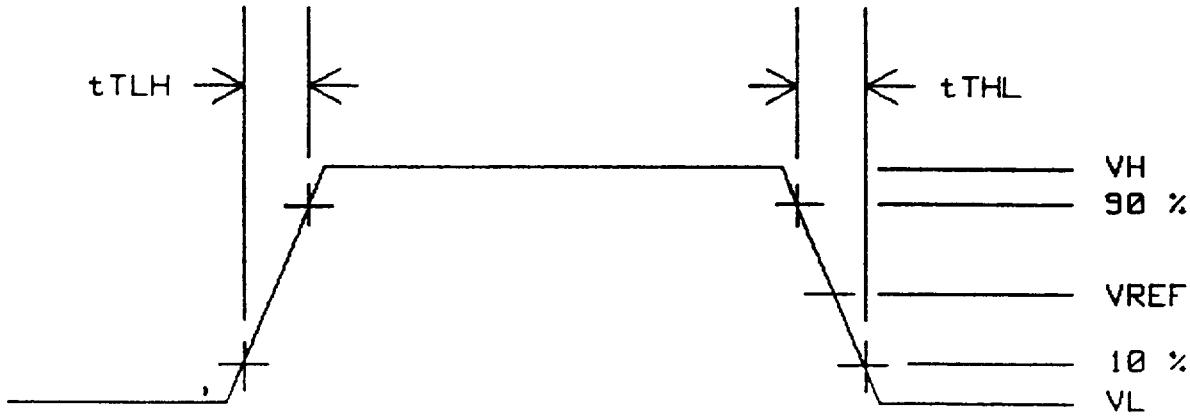
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	COM BUS CLK to COM BUS R0-15	--	--	--	51	51	51	nS	12
2	tpd	COM BUS CLK to ECC ERROR	--	--	--	39	39	39	nS	12
3	tpd	COM BUS CLK to BIT ERROR	--	--	--	59	59	59	nS	12
4	tpd	COM BUS CLK to COM BUS R0-15	--	--	--	86	86	86	nS	12
5	tsu	COM BUS 0-15 to COM BUS CLK	64	64	64	--	--	--	nS	12
6	tpd	COM BUS CLK to MEM DATA 16-21	--	--	--	97	97	97	nS	12

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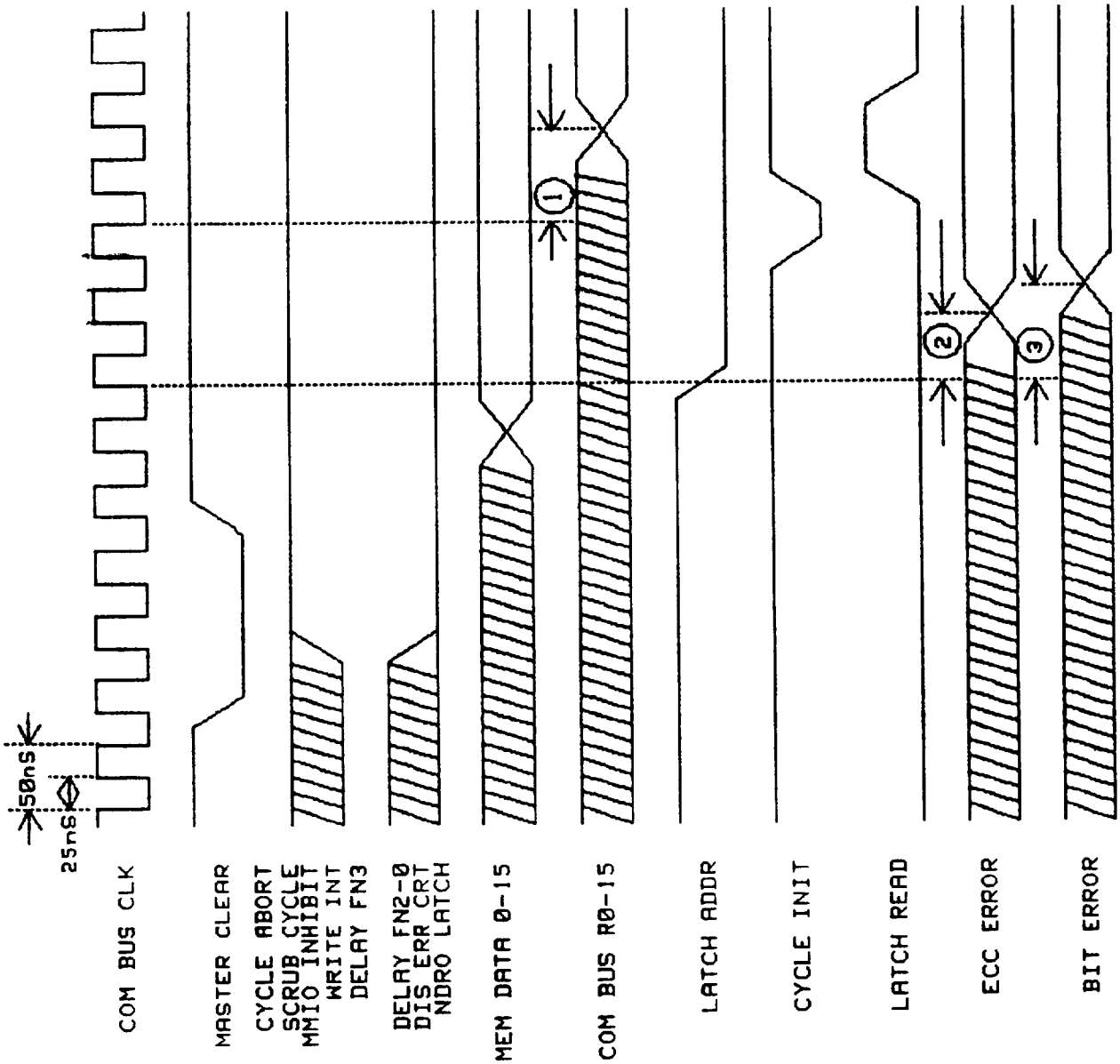


FIGURE 12a. Dynamic waveforms for table XVII.

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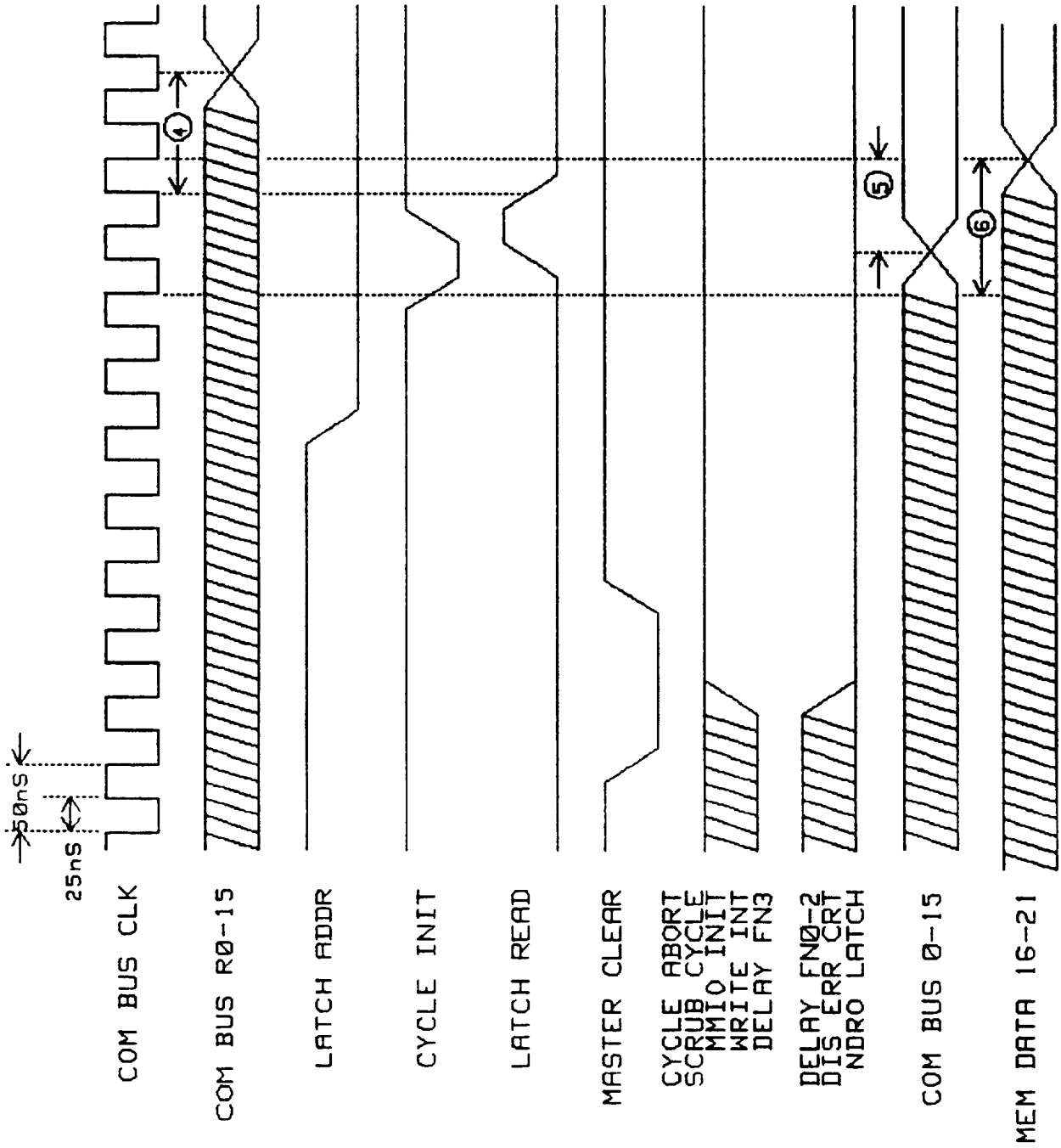


FIGURE 12b. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
	Inputs					
R1	COM BUS 4	30	2	75	pF	1-4
R2	COM BUS 5	32				
R3	COM BUS 6	33				
R4	COM BUS 7	34				
R5	COM BUS 8	35				
R6	COM BUS 9	36				
R7	COM BUS 10	37				
R8	COM BUS 11	38				
R9	COM BUS 12	39				
R10	COM BUS 13	41				
R11	COM BUS 14	42				
R12	COM BUS 15	43	2	75	pF	1-4
R13	TYPE 1	52	4	75	pF	1-4
R14	TYPE 0	53	4	75	pF	1-4
R15	REFRESHCYCLE	54	2	50	pF	1-4
R16	CYCLE ABORT	55	16	150	pF	1-4
R17	CYCLE INIT	56	8	175	pF	1-4
R18	SCRUB CYCLE	57	12	150	pF	1-4
R19	MMIO INHIBIT	58	4	150	pF	1-4
R20	COM BUS CLK	62	2	100	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	DELAY FN2	63	6	100	pF	1-4
R22	DELAY FN1	64	4	75	pF	1-4
R23	SIZE 1	65	2	125	pF	1-4
R24	MEMORY HERE	66	2	75	pF	1-4
R25	STRAP BIT 1	67				
R26	STRAP BIT 2	68				
R27	STRAP BIT 3	69	2	75	pF	1-4
R28	SIZE 0	70	2	100	pF	1-4
R29	HARD ERROR	71	2	75	pF	1-4
R30	SOFTERRORLOG	72				
R31	HARDERRORLOG	73				
R32	DELAY FN0	74	2	75	pF	1-4
R33	LATCH READ	77	2	125	pF	1-4
R34	COM BUS 0	80	2	75	pF	1-4
R35	COM BUS 1	82				
R36	COM BUS 2	84				
R37	COM BUS 3	85	2	75	pF	1-4
R38	MASTER CLEAR	86	4	75	pF	1-4
R39	DIS ERR CRT	89	4	75	pF	1-4
R40	LATCH ADDR	92	6	100	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	LATCH WRITE	93	4	75	pF	1-4
R42	WRITE INT	94	10	150	pF	1-4
R43	NDRO LATCH	95	2	75	pF	1-4
R44	DELAY FN3	97	12	175	pF	1-4
	Bidirectionals					
R45	MEM DATA 0	12	18	200	pF	1-4
R46	MEM DATA 6	13				
R47	MEM DATA 2	17				
R48	MEM DATA 3	18	18	200	pF	1-4
R49	MEM DATA 21	23	12	150	pF	1-4
R50	MEM DATA 16	24	12	150	pF	1-4
R51	MEM DATA 4	26	18	200	pF	1-4
R52	MEM DATA 18	27	12	150	pF	1-4
R53	MEM DATA 19	28	12	150	pF	1-4
R54	MEM DATA 1	29	18	200	pF	1-4
R55	MEM DATA 7	44	16	200	pF	1-4
R56	MEM DATA 9	45	20	200	pF	1-4
R57	MEM DATA 10	46				
R58	MEM DATA 8	47	20	200	pF	1-4
R59	MEM DATA 20	48	12	150	pF	1-4
R60	MEM DATA 5	49	18	175	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R61	MEM DATA 17	50	12	150	pF	1-4
R62	MEM DATA 15	76	18	200	pF	1-4
R63	MEM DATA 13	78	20	200	pF	1-4
R64	MEM DATA 14	79				
R65	MEM DATA 12	87				
R66	MEM DATA 11	88	20	200	pF	1-4
Outputs						
R67	COM BUS R15	2	4	75	pF	1-4
R68	COM BUS R14	3				
R69	COM BUS R13	4				
R70	COM BUS R12	5				
R71	COM BUS R11	6				
R72	COM BUS R10	7				
R73	COM BUS R9	8				
R74	COM BUS R8	9				
R75	COM BUS R7	11				
R76	COM BUS R6	14				
R77	COM BUS R5	15				
R78	COM BUS R4	16				
R79	COM BUS R3	19	4	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R80	COM BUS R2	20	4	75	pF	1-4
R81	COM BUS R1	21	4	75	pF	1-4
R82	COM BUS R0	22	4	100	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R82
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R82
Capacitance measurement	NA	± 3.65 pF	R1-R12
	NA	± 3.55 pF	R13,R14
	NA	± 2.4 pF	R15
	NA	± 6.7 pF	R16
	NA	± 8.35 pF	R17
	NA	± 6.9 pF	R18
	NA	± 7.3 pF	R19
	NA	± 4.9 pF	R20
	NA	± 4.7 pF	R21
	NA	± 3.55 pF	R22
	NA	± 6.15 pF	R23
	NA	± 3.65 pF	R24-R27
	NA	± 4.9 pF	R28
	NA	± 3.65 pF	R29-R32
	NA	± 6.15 pF	R33
	NA	± 3.65 pF	R34-R37
	NA	± 3.55 pF	R38,R39
	NA	± 4.7 pF	R40
	NA	± 3.55 pF	R41
	NA	± 7.0 pF	R42
	NA	± 3.65 pF	R43
	NA	± 8.15 pF	R44
	NA	± 9.1 pF	R45-R48
	NA	± 6.9 pF	R49,R50
	NA	± 9.1 pF	R51
	NA	± 6.9 pF	R52,R53
	NA	± 9.1 pF	R54
	NA	± 9.2 pF	R55
	NA	± 9.0 pF	R56-R58
	NA	± 6.9 pF	R59

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
	NA	± 7.85 pF	R60
	NA	± 6.9 pF	R61
	NA	± 9.1 pF	R62
	NA	± 9.0 pF	R63-R66
	NA	± 3.55 pF	R67-R81
	NA	± 4.8 pF	R82

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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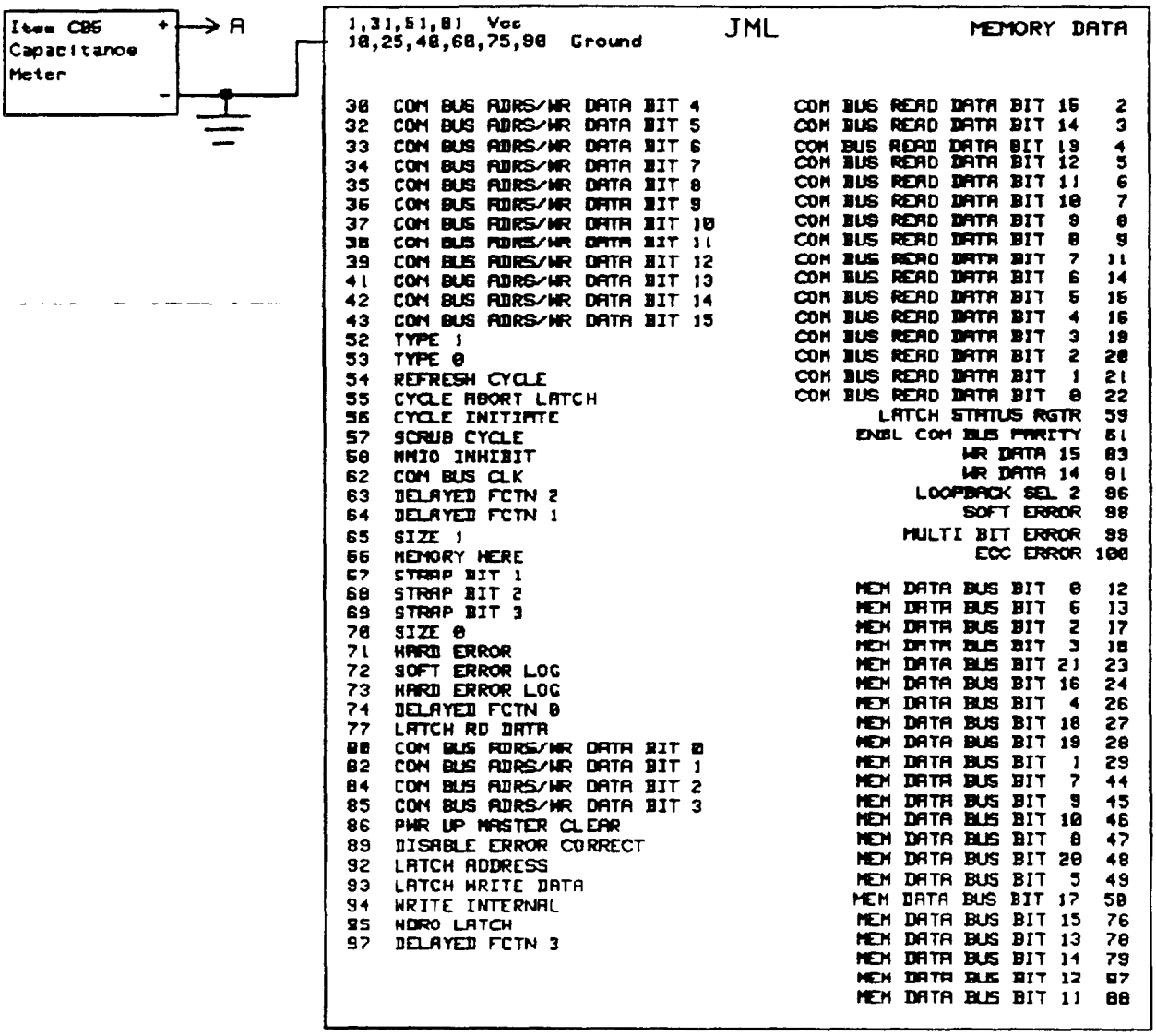


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Input clamp diode latch-up test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
S1	CYCLE INIT	56	-18.0	-1.2	2.4	V	1-3
S2	COM BUS CLK	62	″	″	″	″	″
S3	LATCH WRITE	93	-18.0	-1.2	2.4	V	1-3

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NOTES:

1. Input clamp diode latch-up test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	S1-S3
Power supply current limit	NA	NA	S1-S3
Forcing current, (Iin)	-18 mA	± 1.8 mA	S1-S3
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 S1-S3 S1-S3
Input voltage measurement	NA	± 60 mV	S1-S3

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XIX and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	2.52 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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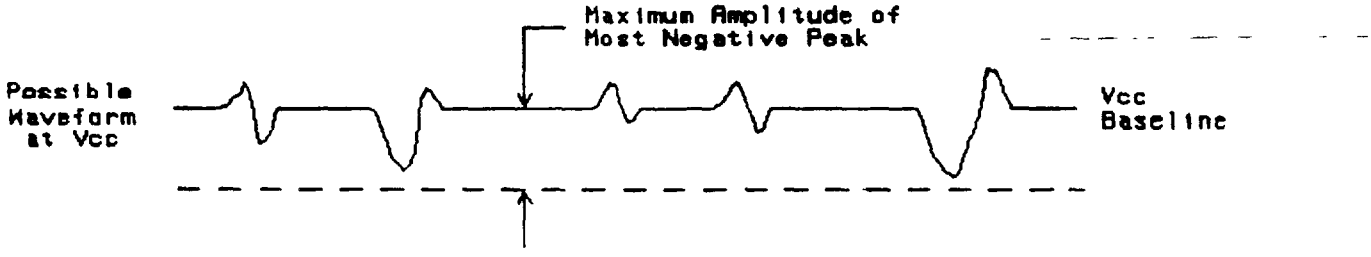
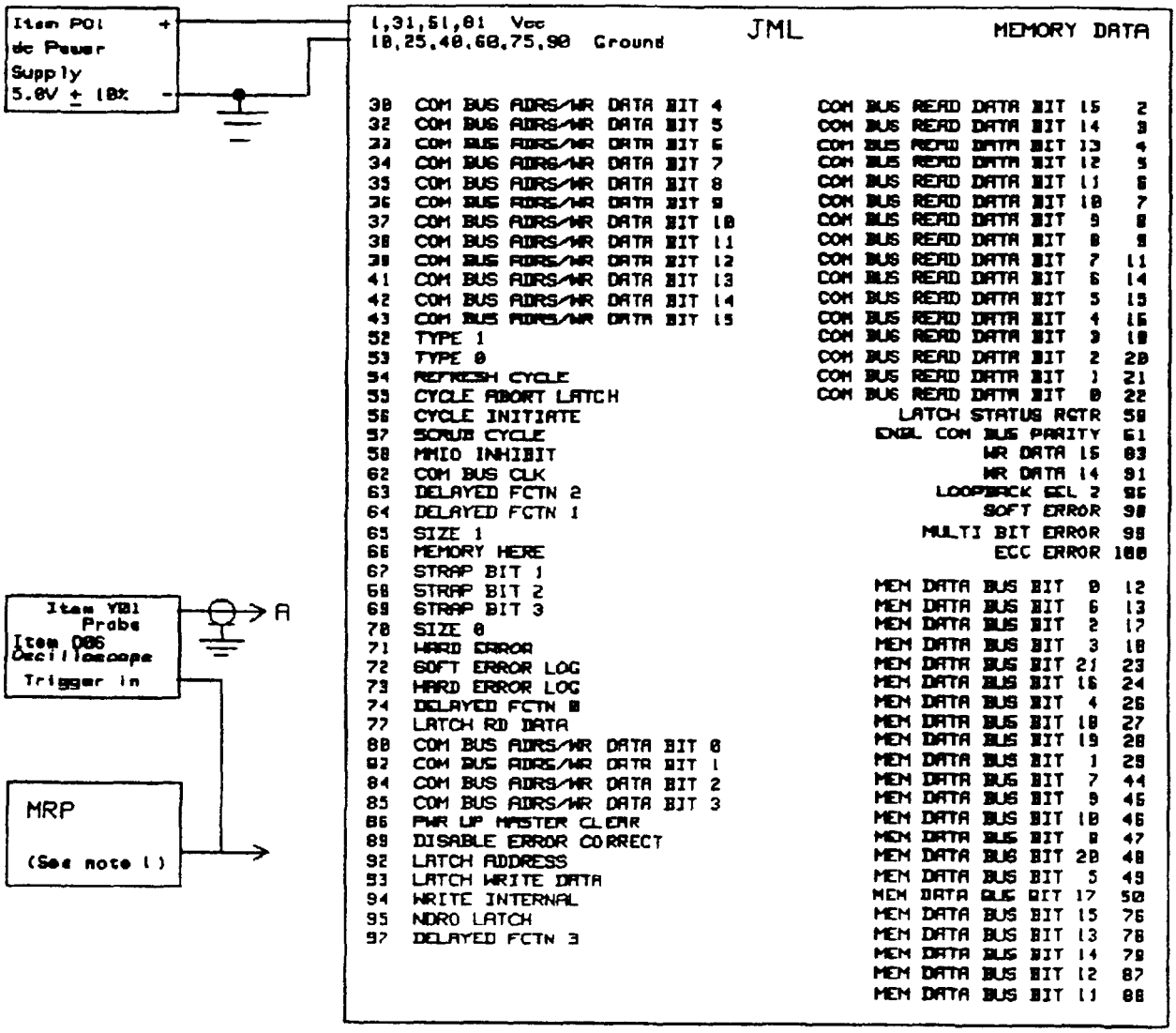


FIGURE 14. Power supply transient amplitude waveform for table XX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XX.

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Modules, Standard Electronic, Memory Data Module, Key Code JML

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